Communicative Efficiency in Adjective Ordering: Plan-Guided Production between Hierarchical Composition and Incremental Pragmatics

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Abstract

Adjective ordering reflects stable order expectations and the usefulness of properties in a referential context. Accounts of communicative efficiency appeal to both hierarchical semantic composition, which interprets noun-adjacent modifiers first, and incremental production, which makes linearly earlier information available first. We examine their contributions through German slider-rating and production studies, Bayesian comparison of Rational Speech Act models, and controlled simulations. Across two slider studies, size-first preference was strongest when size alone identified the target and weakest when colour alone did, with a small size-first preference remaining in the latter context. In production, speakers often selected a single sufficient adjective, while lower size discriminability increased redundant adjective use most clearly when size was sufficient. These findings motivate modelling adjective order together with adjective selection and utterance length. Among models predicting the full distribution of fifteen utterances, plan-guided incremental production provided the best predictions by combining complete-utterance evaluation with competition among successive adjective choices. Allowing the contribution of successive choice to vary across participants further improved prediction of multi-adjective utterances. Context-fixed semantics predicted responses more accurately than sequential context updating, with the difference concentrated in adjective selection and length. Controlled simulations showed that both semantic regimes could yield a size-first target-recovery advantage, depending on how utterances were evaluated and which alternatives were available. Together, the results connect adjective ordering to the joint evaluation of what to mention, how much to say, and which alternatives compete during production.
Keywords: adjective ordering, reference production, Rational Speech Act, incremental production, Bayesian cognitive modelling

1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with scene-specific informativeness in referential communication. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Sproat and Shih, 1991; Scontras, 2023). Yet the first adjective is also the listener's first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

But what counts as efficient—and therefore early—for the listener in a complex referential setting? One relevant line of work concerns adjective subjectivity. More subjective adjectives tend to precede less subjective adjectives in prenominal sequences (Scontras, Degen, and Noah D Goodman, 2017). Some subjectivity-based proposals connect this tendency to hierarchical semantic composition, under which the noun-adjacent adjective composes before adjectives farther from the noun. In *the big blue sticker*, the noun-adjacent adjective *blue* therefore combines with *sticker* before the linearly earlier *big*, narrowing the comparison class against which the gradable size adjective is interpreted. Because *blue* is typically less subjective than *big*, its noun-adjacent position may provide a comparatively stable restriction before the size adjective is interpreted, potentially contributing to the surface size-first preference (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Noah D Goodman, 2019; Scontras, Bar-Sever, et al., 2020). A separate line of referential work suggests that adjective order is also sensitive to the discriminatory strength of each property in the current scene (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). When colour distinguishes the target more effectively than size, colour-first orders may become more acceptable or more likely, consistent with a pressure to provide contextually useful information earlier in the surface sequence.

Depending on which perspective is taken, communicative efficiency may support conflicting adjective-ordering predictions. This tension is especially consequential in reference games (Franke, 2009; Frank and Noah D Goodman, 2012), where successful communication depends both on the completed description and on how information becomes available as that description unfolds. Work on incremental language processing emphasizes the latter: listeners integrate partial linguistic input with visual context to restrict candidate referents before the noun phrase is complete (Tanenhaus et al., 1995; Sedivy et al., 1999). On this view, placing the most discriminating modifier early in the linear sequence may facilitate identification before the remaining words are heard (Rubio-Fernandez and Jara-Ettinger, 2020). On the other hand, the subjectivity-based account emphasizes the former by evaluating a completed adjective sequence through the listener's hierarchical interpretation. From this comprehension-oriented perspective, an order is efficient when the noun-adjacent adjective provides a more reliable restriction for interpreting the outer adjective (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Noah D Goodman, 2019; Scontras, Bar-Sever, et al., 2020).

Moreover, the ordering choices available to a speaker depend on which properties are selected and how much information the description includes. In a colour-sufficient scene, *the blue sticker* may complete the utterance, so competition between size–colour and colour–size orders may never arise. In a size-sufficient scene, speakers may still produce *the big blue sticker*, including colour even when size alone identifies the target. This creates an asymmetry in the redundant use of colour and size adjectives. Previous empirical studies show that colour is included redundantly more often than relative adjectives such as size (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016). Such redundancy can still support efficient identification, especially when modifier meanings are uncertain and gradable, as with size adjectives (Rubio-Fernández, 2016; Degen et al., 2020). Scene-specific usefulness may therefore affect which adjectives are selected and when production stops, as well as how jointly selected adjectives are ordered.

These choices make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome. Within this response space, adjective order is conditional on adjective selection and utterance length. The usefulness of each adjective also depends on the other properties included in the description. This dependence raises a question about production architecture. A global speaker compares complete utterances as wholes. A fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced. Between these endpoints, an utterance-level plan may constrain production while local alternatives continue to influence successive word choices. We call this intermediate hypothesis *plan-guided incremental production*. Under this hypothesis, an utterance-level evaluation combines with the scene-specific usefulness of successive adjective choices. This possibility connects to broader literature on bounded planning, under which speakers can begin speaking before every part of an utterance has been prepared and can continue planning during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014).

We therefore ask three questions, moving from the empirical patterns to computational accounts of the mechanisms that may generate them. First, how do stable size-first preferences and scene-specific referential usefulness jointly shape explicit order judgements and the production outcomes of adjective selection, utterance length, and conditional order? Second, which production architecture best predicts this distribution: global evaluation of complete utterances, fully incremental adjective choice, or plan-guided incremental production between these endpoints? Third, does updating the context for interpreting size improve prediction of the production responses?

To address these questions, we combine carefully controlled behavioural experiments, computational cognitive modelling, simulations across a broader range of randomly generated contexts, and Bayesian predictive model comparison. The empirical programme separates explicit ordering preferences from the production choices that make an order observable. Studies 1a and 1b use a slider-rating task, with Study 1b providing a direct replication in a new sample. Both measure participants' judgements of size-first and colour-first referential utterances across referential contexts. Study 2 uses direct production to elicit utterances and records adjective selection, utterance length, and order conditional on producing the same adjective set. By varying referential context and size discriminability, we separate the relative usefulness of size in a particular scene from its perceptual reliability throughout the task. This manipulation tests the role of perceptual reliability while holding lexical subjectivity constant.

On the modelling side, we formalise these questions within the Rational Speech Act framework (Frank and Noah D Goodman, 2012; Noah D Goodman and Frank, 2016; Degen, 2023). We extend the qualitative model of Schlotterbeck and Wang (2023) from two fixed adjective orders to the full production distribution. That account combined sequential context updating within hierarchical semantic composition with incremental speaker choice. The separate predictive contributions of these two mechanisms remained unresolved. All production models predict the same distribution of one-, two-, and three-adjective responses involving size, colour, and form. The inclusion of form extends the comparison to utterances in which several properties can compete for mention and initial position. They share controls for stable order expectations, visual salience, adjective selection, and utterance length.

Predictive model comparison asks which components improve prediction of the observed production distribution (Vehtari, Gelman, and Gabry, 2017). A separate simulation compares the semantic regimes on the same randomly generated contexts to ask when updating generates an ordering asymmetry and whether that asymmetry also arises under context-fixed semantics. By introducing the production model's utterance evaluation and broader alternatives in controlled steps, it examines how the communicative advantage changes when ordering is considered together with adjective selection and utterance length.

Empirically, the slider studies reveal a replicated context gradient: size-first preferences are strongest when size identifies the target and weakest when colour does. A small residual size-first preference remains when colour alone identifies the target. The production study shows that speakers often resolve reference with one adjective, so adjective selection and stopping frequently determine whether an order contrast arises. When size differences are less distinct, speakers include more adjectives than are needed to identify the target, especially when size alone is sufficient.

After accounting for variation across participants in stable order preferences, the best predictions combine evaluation of complete descriptions with continued influence from successive adjective choices. This pattern is consistent with plan-guided incremental production. Models that interpret size relative to the original scene predict production more accurately than models that update this context as adjective meanings combine. This predictive difference is concentrated in adjective selection and utterance length. In the simulations, a size-first order can give the listener a better chance of identifying the target under both fixed and updating interpretations. This advantage alone therefore does not identify a contribution from semantic updating.

The remainder of the paper is structured as follows. Section 2 develops the theoretical motivation for the experiments and models. Section 3 reports the behavioural studies. Section 4 defines the formal framework for the complete production distribution. Section 5 introduces the matched comparisons, answers the two computational questions in the fitted production data, and examines the best-predicting model. Section 6 evaluates which computational conditions make an ordering advantage depend on semantic updating, and Section 7 considers the implications and limits of the findings.

2 Sources of Adjective Ordering Preferences

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Sweet, 1898; Whorf, 1945; Martin, 1969), class-based hierarchies (Quirk et al., 1972; Dixon, 1982; Cinque, 1994; Scott, 2002), and information-theoretic measures (Hahn et al., 2018; Futrell, Dyer, and Scontras, 2020; Dyer et al., 2023). These predictors capture stable regularities in adjective order, while leaving room for contextual variation (for overviews, see Dyer et al., 2023; Scontras, 2023). Referential context contributes further scene-specific variation in ordering preferences (Fukumura, 2018; Trotzke and Wittenberg, 2019). The present account brings together three lines of work: subjectivity and hierarchical semantic composition, discriminatory strength in reference production, and incremental pragmatic models. Their connection lies in how the usefulness of an adjective depends both on its interpretation and on its contribution to a developing description.

2.1 Subjectivity, Hierarchical Composition, and Adjective Order

An empirical generalisation relates an adjective's *subjectivity* to its preferred distance from the noun: more subjective adjectives are preferred farther from the noun (Scontras, Degen, and Noah D Goodman, 2017). Subsequent work suggests that this relation recurs across languages (Trainin and Shetreet, 2021; Scontras, 2023). The greater subjectivity of size adjectives relative to colour adjectives predicts a size-first preference.

A communicative explanation of this pattern rests on restrictive composition (Scontras, Degen, and Noah D Goodman, 2019). Subjective meanings allow greater divergence between speakers' and listeners' interpretations, which can make reference less successful. Under hierarchical semantic composition, the adjective closest to the noun restricts the context in which an outer adjective is interpreted. Applying a less subjective adjective first can therefore limit the domain over which a more subjective interpretation may diverge. Simulations formalise this proposed benefit and show that sequential restriction can increase the probability of identifying the intended target (Franke, Scontras, and Simonic, 2019). For gradable size adjectives, restriction can also change the comparison class against which size is evaluated. This mechanism of *incremental semantic restriction* connects subjectivity-based ordering to the communicative consequences of hierarchical composition (Scontras, Bar-Sever, et al., 2020). The ordering pressure arises from the interpretation of the combined description and can therefore operate when a speaker evaluates complete utterances as wholes.

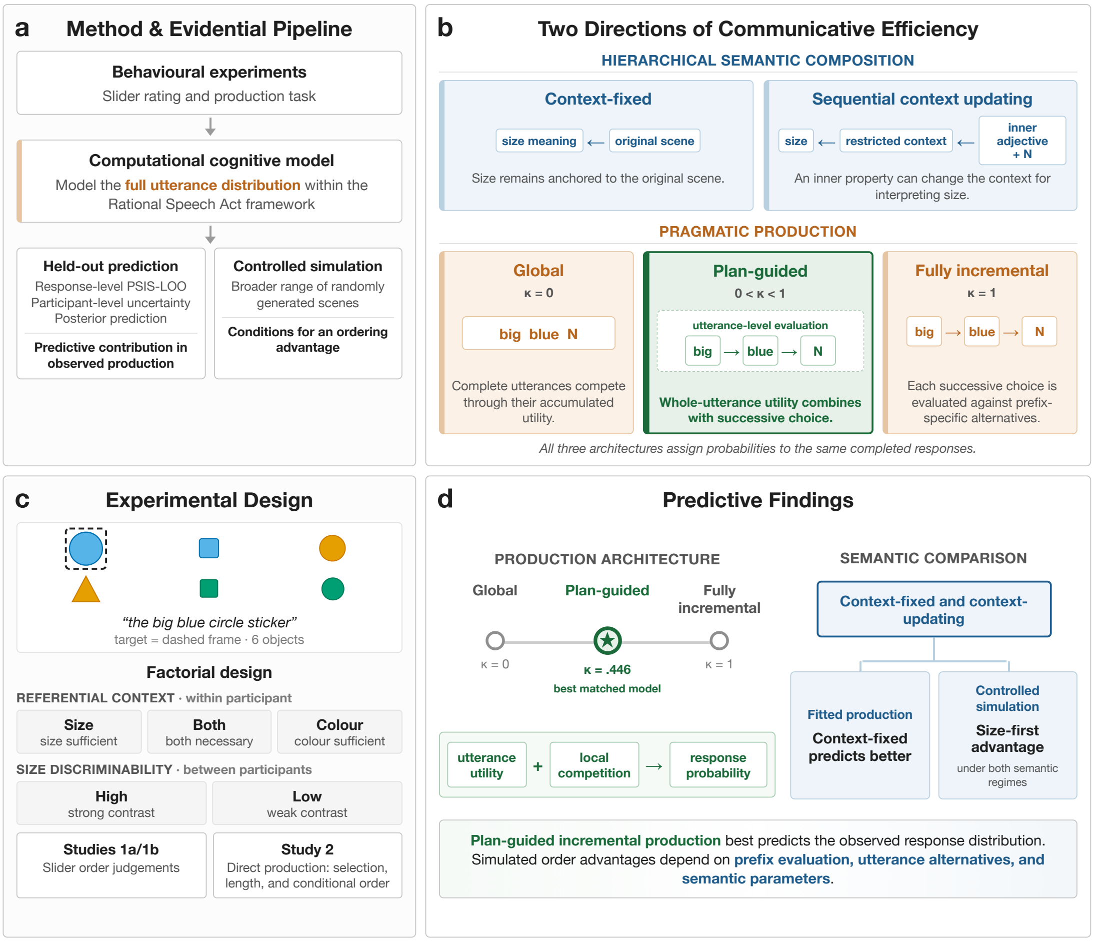


Figure 1: Overview of the design, model space, and evidential structure. Panel A traces the behavioural evidence through computational modelling to held-out prediction and controlled simulation. Panel B distinguishes sequential context updating within hierarchical semantic composition from global, plan-guided, and fully incremental production. Panel C shows the experimental design, and Panel D summarises the predictive result for plan-guided production and the fitted and simulated evidence for sequential context updating.

This account motivates examining whether context updating contributes to the descriptions speakers produce in a particular scene. For size, the relevant comparison is between an interpretation anchored to the original scene and one recomputed as another adjective restricts the context. We refer to these possibilities as *context-fixed* and *context-updating* semantics. Comparing them allows the contribution of semantic updating to be evaluated alongside the stable size-first preference.

2.2 Referential Usefulness, Adjective Selection, and Length

A second ordering pressure comes from referential usefulness in the surface linear sequence. Speakers are more likely to place the more discriminating adjective first, linking ordering to the relative usefulness of properties in the current scene (Fukumura, 2018). The *discriminatory strength* of a property is its ability to distinguish the target from the other objects in the scene. The same adjective can therefore be highly useful in one array and much less useful in another. Under the *incremental communicative efficiency hypothesis*, an early modifier is useful insofar as it narrows the listener's candidate set before the utterance is complete (Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). This proposal accords with evidence that listeners integrate partial linguistic input with visual context (Tanenhaus et al., 1995; Sedivy et al., 1999). The communicative benefit thus depends on when information becomes available in the surface sequence. This connects ordering to the demands of the immediate scene, which may favour an order that departs from the stable size-first preference.

Referential usefulness also acts before an ordering choice becomes observable. Models of reference planning propose that the contrast set shapes which properties are selected for mention (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). The inclusion of redundant material reflects both the processes through which a description is planned and its usefulness to a listener. Speaker-internal accounts of reference production, often termed *egocentric*, propose that readily accessible properties can be selected before the speaker has completed the search for a sufficient description (Pechmann, 1989; Engelhardt, Bailey, and Ferreira, 2006). Audience-oriented accounts assign the additional material a cooperative role: redundant properties can guide visual search and hedge uncertainty about which cue the listener will use (Rubio-Fernández, 2016; Rubio-Fernandez, 2019; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). In a shared visual scene, these routes can converge because a property that is easy for the speaker to retrieve may also help the listener identify the target. These accounts motivate modelling the pressures on content and length alongside adjective order.

Adjective selection and utterance length determine which ordering choices arise in production. A one-adjective utterance can resolve reference without creating an order contrast. Slider ratings hold adjective selection and utterance length fixed by presenting two complete orders. They establish an explicit ordering preference and leave open how often production reaches that choice. Direct production reveals how referential context affects the selection of adjectives as well as their order when selected together. The full distribution of one-, two-, and three-adjective utterances therefore connects ordering preferences to the broader decisions involved in constructing a description.

2.3 Incremental RSA and the Full Production Distribution

The Rational Speech Act (RSA) framework models speaker choice in terms of an utterance's usefulness to a listener and its production cost (Frank and Noah D Goodman, 2012; Noah D Goodman and Frank, 2016; Degen, 2023). It allows assumptions about semantic interpretation and speaker choice to be specified separately within a common account of reference. An ordering preference can consequently depend both on how the adjectives are interpreted and on which alternatives compete during production. Standard RSA models treat complete utterances as the units of speaker choice. The same pragmatic reasoning can be defined incrementally, with the speaker choosing one word at a time relative to the alternatives available after the words already selected (Cohn-Gordon, Noah D. Goodman, and Potts, 2019). The alternatives available at each step can affect adjective selection and the decision to continue or stop, with consequences for the distribution of complete utterances. Pragmatic models of reference capture related asymmetries in modifier position, order, and redundancy (Degen et al., 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Variation in planning scope further motivates considering how complete-utterance evaluation and successive choices may jointly influence production (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). An utterance-level plan may guide production while alternatives available at each step continue to influence the next word. This possibility motivates *plan-guided incremental production*, in which both forms of evaluation contribute to the choice of a completed utterance.

Schlotterbeck and Wang (2023) introduced the slider paradigm on which Studies 1a and 1b build. Their qualitative RSA model combined sequential context updating within hierarchical semantic composition with incremental speaker choice. Together, these mechanisms captured a stable size-first preference and its modulation by discriminatory strength. The analysis considered two fixed size–colour orders and left the separate predictive contributions of the two directions unresolved. The present study extends that starting point with a direct slider replication and a production distribution over one-, two-, and three-adjective utterances, allowing adjective selection, utterance length, and order to be modelled jointly. Evaluating semantic updating and speaker choice separately within this response space allows their contributions to be assessed across content, length, and order. The controlled simulation complements this comparison by examining how the same mechanisms generate ordering advantages across a broader range of scenes.

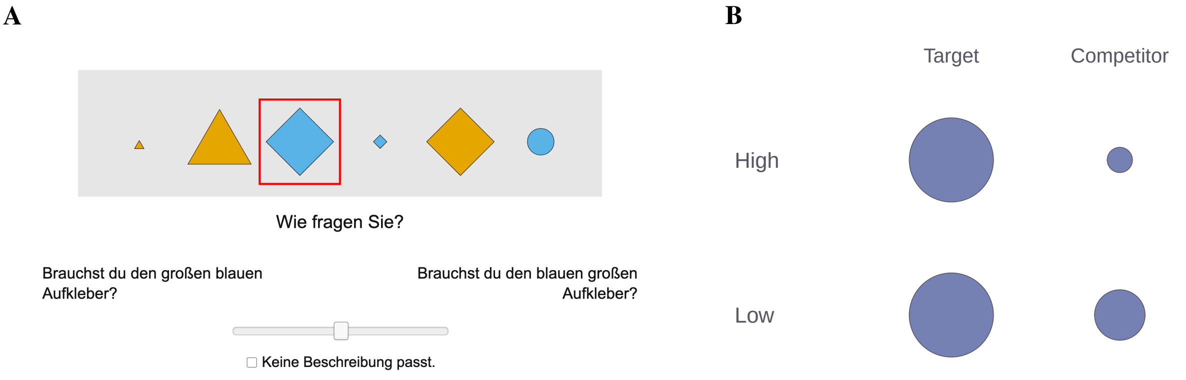
3 Behavioural Experiments

The three studies expose different aspects of the same empirical problem. Study 1a and its direct replication, Study 1b, ask participants to compare two completed size–colour orders using sliders. Study 2 elicits descriptions through direct production, allowing adjective selection and utterance length to vary along with order. The two tasks therefore connect explicit ordering preferences to the choices that make an order observable in production. Referential context varies whether size, colour, or both are needed to identify the target, manipulating the relative usefulness of the two properties (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Size discriminability separately varies how clearly size differences can be perceived. Crossing these factors allows us to ask whether the influence of perceptual reliability depends on which properties support reference, and how these pressures affect ordering judgements and production. Because size is gradable, this comparison also bears on the role of the comparison class in size interpretation (Scontras, Degen, and Noah D Goodman, 2019; Schlotterbeck and Wang, 2023).

3.1 General Method Shared between Experiments

3.1.1 Design

The three studies shared the same basic factorial design and visual displays, with different response formats. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. Referential context varied within participants: size alone, colour alone, or both properties were needed to identify the target. Size discriminability varied between participants, so that each participant experienced either more pronounced or less pronounced target–competitor size differences throughout the task (Figure 2). The behavioural comparisons below focus on size–colour trials. The production study also included size–form and colour–form trials, which enter the later computational model comparison.



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Design dimension	Levels
Referential context	Size sufficient; both necessary; colour sufficient (within participants)
Size discriminability	High; low (between participants)

Figure 2: Experimental design and materials. Panel A reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders. Panel B illustrates the size-discriminability manipulation schematically using recorded stimulus levels: the target–competitor size gap is larger under high discriminability and smaller under low discriminability. Panel C summarises the factorial design for the size–colour comparisons.

3.1.2 Materials and procedure

All studies were conducted online via Prolific. Compensation was set slightly above the platform's median level. The task was framed as a reference game about collectible stickers: participants gave or evaluated a description for a partner who saw the same stickers in a different order and lacked the target frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. The noun was always *Aufkleber* ('sticker'), and the materials varied the objects' size, colour, and form across counterbalanced lists. Practice trials familiarised participants with the task, and main-trial order was randomised within blocks. Stimulus construction, trial counts, exclusions, and coding details appear in Appendix A.

3.2 Study 1a: Slider-Rating Study

3.2.1 Participants

Study 1a uses the same dataset reported by Schlotterbeck and Wang (2023). The retained sample comprised 134 participants (70 in the low- and 64 in the high-discriminability group), recruited via Prolific and reporting German as their native language. Exclusions addressed missing records, extreme completion times, and failures on control or task-relevance checks. The size–colour analysis contained 3,485 observations.

3.2.2 Procedure

On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation of the two orders was randomised across trials. Ratings were centred so that zero indicated neutrality, positive values favoured size-first, and negative values favoured colour-first order, on a scale from -0.5 to 0.5.

3.2.3 Results

The influence of size discriminability on ordering judgements varied with referential context (Figure 3). The clearest difference was between size-sufficient and both-necessary displays: higher discriminability shifted the estimated ratings towards size-first in the former and towards colour-first in the latter. The planned contrast supported this difference between effects ($-.085$, 95% credible interval $[-.151, -.022]$), although the individual high–low differences remained uncertain. Evidence for the corresponding interaction involving colour-sufficient displays was less clear. Study 1b tests whether this interaction pattern recurs.

Across discriminability conditions, ratings followed a clear referential-context gradient. Size-first preferences were strongest when size alone identified the target, weaker when both properties were needed, and weakest when colour alone identified it. Even in colour-sufficient displays, the estimated preference remained slightly size-first. Thus, referential usefulness modulated the strength of the ordering preference without producing an overall colour-first reversal. These conclusions were evaluated with a Bayesian hierarchical model containing context, discriminability, and their interaction. Model specifications appear in Appendix A.

3.3 Study 1b: Direct Slider Replication

3.3.1 Participants and procedure

The replication used the same task and materials with an independent participant sample. Applying the Study 1a exclusion criteria retained 104 of 132 participants and 2,786 size–colour observations.

3.3.2 Results

The replication did not provide clear evidence for the interaction pattern observed in Study 1a (Figure 3). Under the same Bayesian hierarchical model, both planned interaction contrasts remained uncertain. Higher size discriminability was associated with higher estimated size-first ratings in all three contexts, with the clearest evidence in size-sufficient and colour-sufficient displays. The estimated directions therefore differed from the crossover in Study 1a.

The referential-context gradient was replicated: size-first preferences were strongest in size-sufficient displays, intermediate when both properties were needed, and weakest in colour-sufficient displays. A small positive size-first preference also remained when colour alone identified the target. Together, the slider studies provide consistent evidence that referential usefulness modulates explicit ordering preferences, while the context-dependent influence of size discriminability was less consistent across studies.

3.4 Study 2: Production Study

3.4.1 Participants

The production study retained 113 participants (60 in the low- and 53 in the high-discriminability group), recruited via Prolific; participants from either slider study were ineligible. Exclusions addressed extreme completion times, missing or unannotatable responses, and more than 80% identical responses. The latter criterion selects a sample that varies its descriptions across trials. The behavioural comparisons use 3,034 size–colour trials; the later computational comparison uses all 9,100 annotated trials across the three adjective families.

3.4.2 Procedure and outcome measures

Participants saw the framed target together with the fixed sentence frame *den ___ Aufkleber* and typed the adjective material before the noun. Practice trials introduced the relevant colour and form vocabulary. Responses were coded by the first author into 15 categories preserving adjective content and order: three one-adjective, six two-adjective, and six three-adjective strings. The coding dictionary and treatment of unclassifiable responses appear in Appendix A.

Figure 3: Ordering judgements in Study 1a and Study 1b by referential context and size discriminability. Points show posterior medians of expected centred ratings for a new participant; error bars show 95% credible intervals. Positive values indicate size-first preference; zero indicates neutrality. The two studies use the same model specification and interval definition.

Two outcomes connect production to the experimental hypotheses. *Redundant adjective use* indicates a response containing more properties than needed for unique reference: multiple adjectives in a one-property-sufficient context, or three adjectives when both properties are needed. *Size-initiality* indicates that a response begins with size, including a response consisting of size alone. It therefore reflects adjective selection as well as order. Order conditional on producing both size and colour is described separately. The two outcomes were analysed with Bayesian hierarchical models containing referential context, size discriminability, and their interaction; the full specifications and planned contrasts appear in Appendix A.

3.4.3 Results

Redundant adjective use. The influence of size discriminability on redundancy depended on referential context (Figure 5A). Lower discriminability increased redundant adjective use most clearly when size alone was sufficient to identify the target. This increase was larger than the average increase in the other two contexts, by an estimated 15.3 percentage points (95% credible interval $[7.2, 23.6]$). When size differences were less distinct, speakers were therefore particularly likely to supplement an otherwise sufficient size description with additional properties.

The response distribution also shows how referential context shaped content and length more generally (Figure 4). Both-necessary scenes generally elicited longer descriptions, whereas one-property-sufficient scenes more often elicited a single adjective. Colour-sufficient scenes were commonly resolved with colour alone, so no size–colour ordering decision was observed on those trials. These patterns connect the usefulness of a property to both its selection and the amount of material produced.

Initial adjective and conditional order. The influence of size discriminability on size-initiality also depended on referential context (Figure 5B). Higher discriminability increased size-initial responses most clearly when both size and colour were needed. The increase exceeded the average high–low difference in the two one-property-sufficient contexts by 16.9 percentage points (95% credible interval $[7.7, 25.6]$). The corresponding effects were small in size-sufficient displays and negative in colour-sufficient displays. Thus, making size more distinct particularly favoured beginning with size when the description needed both properties.

Conditional on producing both size and colour, size-first descriptions remained more frequent than colour-first descriptions; colour-before-size reversals were comparatively rare. This conditional preference describes responses containing both properties, while size-initiality also includes the decision to use size alone. The production results therefore locate context sensitivity in adjective selection, length, and initial position, alongside a descriptive size-first preference among responses containing both adjectives.

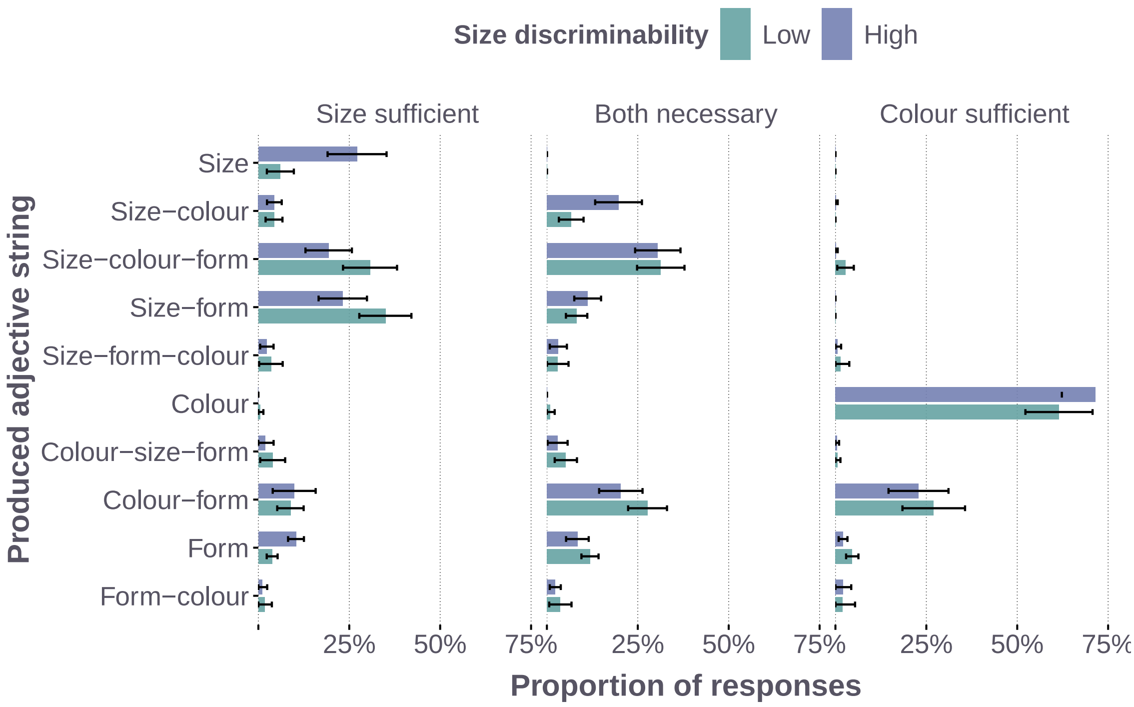

Figure 4: Ordered response distribution across the 3,034 size–colour production trials. Bars show mean participant-level proportions by referential context and size discriminability, with 95% confidence intervals. Adjective labels preserve surface order; the figure shows responses reaching at least 2% in one condition.

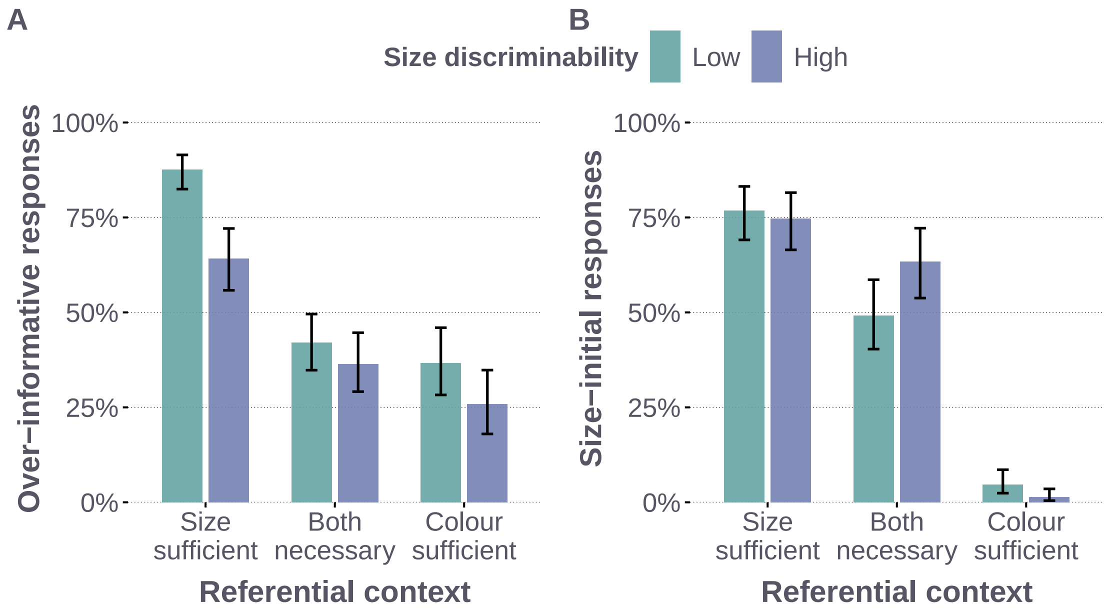

Figure 5: Hypothesis-linked recodings of the size–colour production responses. Panel A codes redundant adjective use as over-informative: a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel B codes whether the response begins with size. Bars show Bayesian posterior medians and error bars show 95% credible intervals. Size-initiality combines adjective selection and order.

3.5 Interim Discussion

The studies provided different patterns of evidence for the interaction between size discriminability and referential context. Study 1a showed the clearest interaction between size-sufficient and both-necessary displays, while Study 1b did not provide clear evidence for the same pattern. In production, lower size discriminability increased redundant adjective use most clearly when size alone was sufficient; higher discriminability increased size-initial responses most clearly when both properties were necessary. Perceptual reliability therefore affected different aspects of a description depending on the referential demands of the scene.

The two slider studies consistently showed a referential-context gradient: size-first preferences were strongest when size identified the target and weakest when colour did, with a small positive preference remaining in colour-sufficient displays. Direct production revealed how the same contextual demands also shaped adjective selection and utterance length. When one property identified the target, speakers frequently produced a single adjective, leaving no order among multiple adjectives to observe. The context manipulation can therefore change explicit judgements between completed orders and the likelihood that production includes those adjectives together.

Production also revealed substantial use of three-adjective utterances and pronounced participant variation in how often these longer descriptions occurred. These responses extend beyond the material required for unique reference and involve several successive adjective choices. Together, the contextual effects and participant variation motivate modelling the complete production distribution, so that selection, length, and order can be evaluated within a common account.

4 Formal Modelling Framework

The formal framework of the current paper builds on the central intuition of the Rational Speech Act (RSA) model: rational communication links comprehension and production, such that a listener interprets an utterance to infer the speaker's intended referent and a pragmatic speaker assigns greater probability to utterances that guide the listener successfully to the target (Frank and Noah D Goodman, 2012). In the vanilla RSA model, the speaker evaluates a fixed set of completed utterances according to their communicative efficiency. The present model extends the vanilla RSA model in three respects. First, it expands the response space to all fifteen ordered one-, two-, and three-adjective utterances available in the experiment, bringing adjective selection, utterance length, and ordering into a single production distribution. This extends the two fixed size–colour orders modelled by Schlotterbeck and Wang (2023) to the richer structure of adjective use observed in direct production. Second, it allows recursive updating of the listener state as adjectival meanings compose from the noun outwards. Third, it extends the speaker to incremental production by allowing adjective choices to be evaluated successively along the surface linear sequence.

These extensions allow the contributions of semantic interpretation and successive adjective choice to be evaluated within the complete production distribution. The production architecture determines how strongly competition among successive adjective choices contributes to the probability of the completed utterance, ranging from global evaluation through plan-guided production to fully incremental adjective choice. The semantic regime determines whether interpreting an adjective can change the comparison class for subsequent hierarchical composition.

4.1 Utterance Space, State Space, and the Literal Listener

The model assigns probabilities to all fifteen ordered strings formed from size, colour, and form: three one-adjective utterances, six two-adjective utterances, and six three-adjective utterances. This response space allows the model to predict which properties speakers mention, how many adjectives they produce, and how jointly selected adjectives are ordered. All models use the observed six-object display. Size has a graded meaning defined relative to a comparison class (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019). Colour provides a graded target-matching cue; form has a neutral literal meaning, and its use is supported through salience and task-specific production terms. For every complete or partial adjective string, the literal listener L applies adjectival modification from the noun outwards, corresponding to right-to-left composition in the prenominal strings considered here.

Let $u_{\leq t} = w_1 \dots w_t$ be a surface prefix comprising the first t adjectives, and $P_0(o \mid X) = 1/6$ the uniform prior over the six objects in display X. Under context-fixed semantics, the literal listener after this prefix is

$$L(o \mid u_{\leq t}, X) \propto P_0(o \mid X) \prod_{j=1}^{t} [\![w_j]\!]_{\text{fix}}(o). \tag{1}$$

The listener combines the meanings made available by the surface prefix and assigns probability to every object that remains compatible with them. The probability assigned to the target provides the communicative value used by all three production accounts.

4.2 Successive Adjective Choice and Plan-Guided Production

The speaker evaluates how the adjectives in an utterance guide the listener to the target at successive positions. At position t, the utility of candidate adjective a for target o^* is

$$U_t(a \mid u_{<t}, o^*, X) = \alpha \log L(o^* \mid u_{<t} + a, X) + \lambda_{\text{salience}} z_a, \tag{2}$$

where α controls sensitivity to listener success and z_a represents the visual salience of the candidate property. For a completed utterance u, the model records both the accumulated utility of its selected adjectives and the competition they faced at successive positions:

$$U^{\text{complete}}(u, o^*, X) = \sum_t U_t(w_t \mid u_{<t}, o^*, X), \tag{3}$$

$$\mathcal{N}^{\text{local}}(u, o^*, X) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}, o^*, X), \tag{4}$$

where $\mathcal{A}(u_{<t})$ contains the adjective types still available after the current prefix. The complete-utterance utility U^{complete} accumulates the utilities of the adjectives selected along the utterance. The local-normalisation term $\mathcal{N}^{\text{local}}$ accumulates the log denominators of the local softmax choices over adjectives available at each position. In a locally normalised choice, a selected adjective receives lower probability when several alternatives at the same prefix also have high utility. Two completed utterances with similar U^{complete} can therefore receive different probabilities because they pass through prefixes with different degrees of competition.

The probability of a completed utterance is

$$S_\kappa(u \mid o^*, X) \propto \exp\bigl\{U^{\text{complete}}(u, o^*, X) - \kappa \mathcal{N}^{\text{local}}(u, o^*, X) + \beta\Omega(u) + T(u, X)\bigr\}, \tag{5}$$

The parameter κ determines how strongly competition at successive adjective positions contributes to the probability of a completed utterance. With $\kappa = 0$, the local normalisers drop out and the *global* speaker normalises over the fifteen completed paths. With $\kappa = 1$, the *fully incremental* speaker retains the complete contribution of locally normalised adjective choices. An interior value defines the *plan-guided* speaker: complete-utterance utility and competition among locally available adjectives jointly determine the response probability. The global model therefore compares complete utterances using their accumulated prefix utilities.

The remaining terms capture stable order expectations through $\beta\Omega(u)$ and task-specific pressures on adjective selection and utterance length through $T(u, X)$. They extend the informativity–cost trade-off of a vanilla RSA speaker: target-directed utility is balanced against conventional order preferences and task-specific production costs. These terms are shared across the three production accounts. Length choice enters through the comparison of completed responses and $T(u, X)$.

Before the final Laplace smoothing, the same model can be written at common parameter values as a geometric interpolation between the global and fully incremental endpoints:

$$S_\kappa(u \mid o^*, X) \propto S_{\text{G}}(u \mid o^*, X)^{1-\kappa} S_{\text{I}}(u \mid o^*, X)^{\kappa}, \tag{6}$$

where S_{G} and S_{I} are the endpoint distributions evaluated with the same listener, stable-order component, task link, and parameter values. The proportionality in Equation (6) renormalises the interpolated scores over the fifteen completed utterances.

The full parameterisation and response likelihood appear in Appendix B; Appendix F derives the local-normalisation term and its relation to the model endpoints.

4.3 Sequential Context Updating within Hierarchical Semantic Composition

The semantic comparison concerns the inner-to-outer direction of hierarchical composition. Let $c_1, \ldots, c_T$ denote the modifiers in a completed utterance ordered from the noun outwards, so that $c_1 = w_T$ for prenominal modification. Starting from $C_0 = P_0$, composition successively restricts the context state available to the literal listener:

$$C_j(o) \propto C_{j-1}(o)\, [\![c_j]\!]_{\rho_j}(o), \qquad j = 1, \ldots, T, \tag{7}$$

where $[\![c_j]\!]_{\rho_j}$ is the modifier's meaning under the selected semantic regime.

The semantic regime determines the comparison class used for size. Under context-fixed semantics, the graded size threshold remains anchored to the original scene. Under sequential context updating, the threshold is recomputed from the context state left by the hierarchically prior modifier. When this restriction makes the target more distinctive in size, sequential context updating can give the size-first order a referential advantage. The literal listener applies this noun-outward computation to each available surface prefix, linking semantic interpretation to the successive adjective utilities defined above. The comparison therefore asks whether recomputing the size comparison class adds predictive accuracy in the observed production data. The simulation examines how this operation contributes to an ordering advantage across production models and a broader range of contexts. The threshold construction and further semantic details appear in Appendices B and G.

5 Model Comparison and Predictive Evidence

The model comparison tests how hierarchical semantic composition and successive adjective choice contribute to the observed distribution of utterances. Global, plan-guided, and fully incremental production are each paired with context-fixed semantics or sequential context updating in a matched 3×2 comparison. We then assess whether participant variation in successive adjective choice improves prediction and examine which production patterns the resulting model captures.

5.1 Inference and Model Comparison Setup

We fit each model to the same 9,100 production trials from 113 participants using Bayesian inference in NumPyro (Phan, Pradhan, and Jankowiak, 2019). All six models share the same fifteen-utterance response space, task link, semantic parameters, and participant hierarchies for pragmatic optimality α_i and stable-order weight β_i. They differ in production architecture, specified by fixing $\kappa = 0$, estimating one shared κ, or fixing $\kappa = 1$, and in whether the interpretation of size is recomputed during hierarchical composition. These controls isolate the predictive contribution of each proposed direction of communicative efficiency. The subsequent participant refinement estimates individual successive-choice weights κ_i within the plan-guided architecture.

Approximate leave-one-out expected log predictive density (ELPD) measures how much probability each model assigns to held-out responses (Vehtari, Gelman, and Gabry, 2017). Higher ELPD indicates better prediction of the complete utterance distribution. The predictive target is an additional completed response from a sampled participant, with that participant's remaining trials available to estimate their participant-specific parameters. Uncertainty in paired model comparisons is estimated with a participant-level Bayesian bootstrap and reported as 95% credible intervals. The same predictive target and uncertainty calculation are used throughout the architecture and hierarchy comparisons. The model specification, inference procedure, and numerical diagnostics appear in Appendices B and C.

5.2 Model-Comparison Results

Production architecture. The plan-guided model predicted production better than either the global or fully incremental model, supporting a joint contribution of complete-utterance evaluation and competition among successive adjective choices. Across semantic regimes, it gained 256.9 ELPD over the global model (95% credible interval $[71.8, 404.7]$) and 548.5 over the fully incremental model ($[349.8, 795.1]$; Figure 6). The estimated shared successive-choice weight was intermediate between the two endpoints ($\kappa = .446$).

The largest differences among architecture predictions occurred in form-involving trials, where colour and form competed for the initial position. Colour-initial responses contributed most of the plan-guided model's advantage over the average endpoint. Different opening adjectives create different sets of plausible continuations, so competition at later positions redistributes probability among the completed utterances that share each initial adjective. The broader production response space thus supplies evidence about successive adjective choice beyond the motivating size–colour order contrast.

Hierarchical semantic composition. Context-fixed semantics predicted the complete response distribution better than sequential context updating under all three production architectures. Updating minus fixed semantics averaged -43.527 ELPD, with a 95% credible interval of $[-56.005, -31.770]$. This preference for context-fixed semantics remained in the participant-specific refinement described below. For that model pair, decomposing predictive accuracy into length, adjective content, and order showed that the disadvantage of updating was concentrated in length and adjective content; the conditional-order contrast remained uncertain. The decomposition therefore locates the semantic difference primarily in how much speakers say and which properties they mention. Full contrasts and the decomposition appear in Appendix C.3.

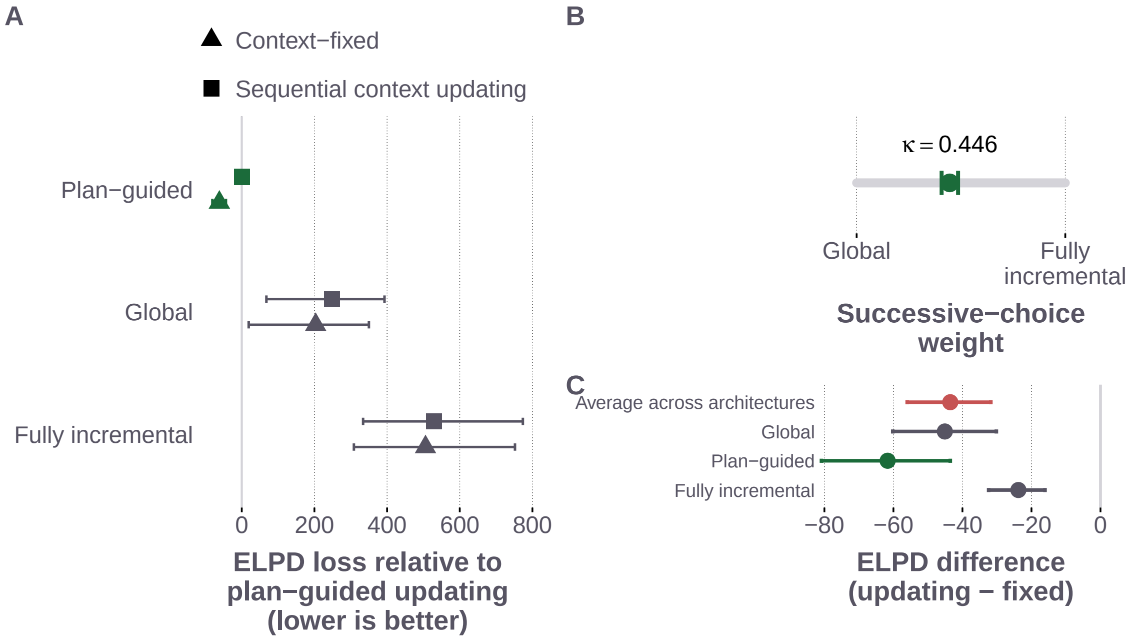

Figure 6: Matched comparisons of production architecture and sequential context updating. Panel A reports predictive loss for all six matched models relative to the plan-guided model with sequential context updating; lower values indicate better leave-one-out prediction. Panel B places the shared successive-choice weight κ between the global and fully incremental endpoints. Panel C shows updating-minus-fixed ELPD for each production architecture and their average. Intervals in Panels A and C are 95% credible intervals from the participant-level Bayesian bootstrap; Panel B shows the 95% posterior credible interval for κ. The vertical line in Panel C marks zero.

5.3 Plan-Guided Production and Participant Variation

Allowing the contribution of successive adjective choice to vary across participants further improved prediction. The refinement retains the existing hierarchies for pragmatic optimality and stable-order weight while replacing the shared κ with participant-specific weights κ_i. It gained 364.742 ELPD under context-fixed semantics, with a 95% credible interval of $[248.178, 520.967]$; the gain under sequential context updating was similar. The population location of the successive-choice weights remained intermediate between the global and fully incremental endpoints, with a broad distribution across participants.

The hierarchy improved prediction for two- and three-adjective utterances, with a smaller loss for single-adjective responses. Its contribution is therefore concentrated in responses containing several adjectives, where successive alternatives can redistribute probability among completed utterances. The gain on three-adjective utterances also shows that participant variation in successive adjective choice helps locate redundant descriptions in the observed distribution. Appendix E reports the response-length contrasts and participant-level diagnostics.

5.4 Predictive Adequacy of the Best-Fitted Model

The plan-guided model with context-fixed semantics and participant-specific successive-choice weights had the highest ELPD. Its posterior predictions captured the broad shifts in size-initial responses and redundant adjective use across the size–colour referential contexts (Figure 7, Panel A). Panel B extends this check to all fifteen completed utterances across the three adjective families, referential contexts, and size-discriminability conditions.

A remaining discrepancy concerns high-discriminability size–form displays where both properties are necessary. Size-initial responses occurred on 59.5% of trials, compared with a prediction of 45.2%, while form-initial responses occurred on 4.2% of trials, compared with 13.5% predicted. The model therefore underpredicts size-initial responses and overpredicts form-initial responses in this condition. Additional response-level checks and descriptive measures of fit appear in Appendix C.

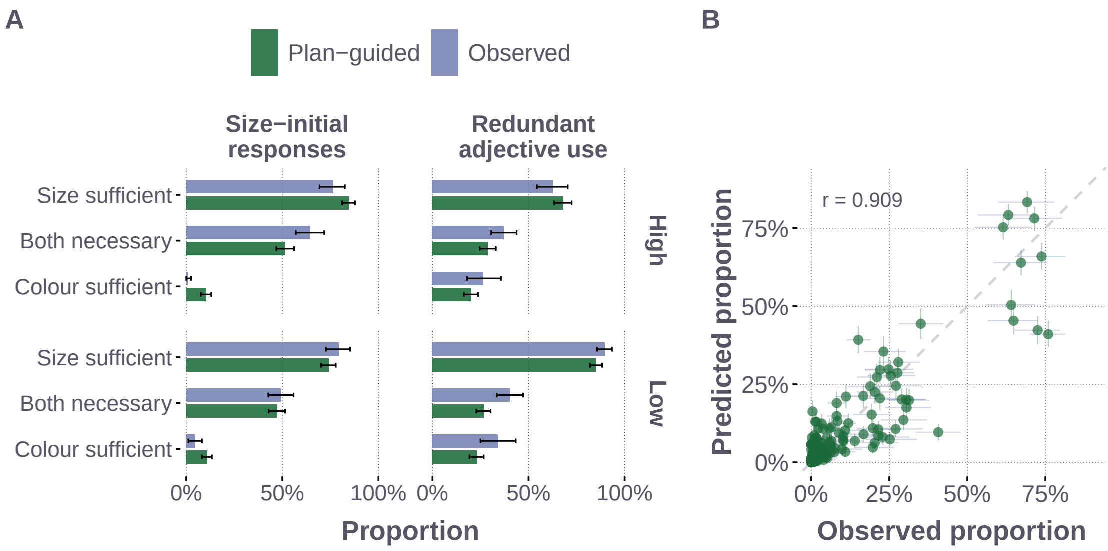

Figure 7: Posterior-predictive adequacy of the best-fitted plan-guided model. Panel A compares observed and posterior-predicted size-initial responses and redundant adjective use in size–colour trials, with referential context on the vertical axis and size discriminability in facet rows. Intervals show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions. Panel B compares all 270 response cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality.

These predictive comparisons leave open how a communicative advantage for size-first order arises under each semantic regime. The following simulation examines how this advantage arises and how it changes when adjective selection and utterance length enter the speaker's alternatives.

6 Simulation and Generative Conditions for an Order Asymmetry

A communicative account of adjective ordering seeks to explain why one order can be more useful for identifying a referent than another. Earlier Monte Carlo simulations found an average referential advantage for orders that place more subjective adjectives farther from the noun, motivating a pressure that could gradually stabilise ordering preferences across repeated language use (Franke, Scontras, and Simonic, 2019). Sequential context updating provides a mechanism for this advantage: the noun-adjacent adjective restricts the context in which an outer, more subjective adjective is interpreted. The present production findings place this proposal within a broader set of choices, because speakers also decide which adjectives to include and how much to say. The theoretical question is therefore how the proposed ordering advantage changes when the speaker evaluates successive adjective information and can choose descriptions of different lengths.

The simulation addresses this question through controlled changes on the same randomly generated reference scenes. It begins with a comparison of the two size–colour orders based on their completed meanings, then introduces the utterance evaluation used by the production model and expands the alternatives to include single adjectives and the full production utterance space. The original scene distribution and listener target-recovery measure remain fixed throughout. This comparison identifies the conditions under which the advantage depends on semantic updating and those under which it also arises with fixed meanings. Although the symmetry of fixed meanings in the completed-utterance baseline can be derived directly, the simulation establishes the average direction and magnitude of the advantage as these production assumptions change across scenes.

6.1 Simulation Design and Size-First Advantage

The simulation uses 24,000 randomly generated displays that vary in the number of objects and their size distribution. The largest object is designated as the target and assigned the colour blue, ensuring that both size and colour apply. The number of blue distractors varies across displays, so colour restricts the comparison class to subsets of different sizes. This extends the analysis beyond the experiment's controlled sufficiency conditions and varies how strongly colour-based restriction can affect the interpretation of size. Greater size spread also tends to separate the largest target more clearly from its competitors, providing a simulation analogue of the experimental size-discriminability manipulation. The same displays are reused across all comparisons. The scene-generation procedure and semantic parameter grid appear in Appendix H.

The semantic regimes follow the distinction in Section 4: size is interpreted relative to the full scene under context-fixed semantics and relative to the context established by inner modifiers under sequential context updating. The initial baseline chooses between *big blue* and *blue big* using only the literal listener's target probability after the completed utterance. The next comparison uses the global production model ($\kappa = 0$), which accumulates listener log probabilities over the successive adjectives before choosing among completed utterances. For *big blue*, this includes the information supplied by *big* as well as the completed description. Both comparisons use global choice; the difference is which information contributes to the utterance's utility. Holding the choice architecture fixed isolates the contribution of successive adjective information.

The global model is then given four alternatives by adding *big* and *blue*, followed by the fifteen one-, two-, and three-adjective utterances used in production. These changes allow the speaker to select a sufficient single adjective or include other properties. The first four comparisons average over the same grid of 125 semantic parameter settings. The final comparison retains the fifteen utterances and substitutes the semantic constants used in production. Pragmatic optimality remains $\alpha = 1$, with order, salience, and task-specific terms set to zero and final Laplace smoothing omitted. The additional architecture comparisons and full computational specification appear in Appendix H.1.

The pragmatic listener L^{prag} applies Bayes' rule to the speaker distribution: an object receives more probability when a speaker intending that object would be more likely to produce the observed utterance. All objects have equal prior probability. For each context, the size-first advantage compares the listener's target probability after the two orders:

$$\Delta_{\text{SF}} = L^{\text{prag}}(o^* \mid \text{“big blue”}) - L^{\text{prag}}(o^* \mid \text{“blue big”}). \tag{8}$$

Positive values indicate better pragmatic target recovery for the size-first order. Because the listener reasons about the speaker's choice, other available descriptions can affect this measure even though the two utterances being evaluated remain the same.

6.2 Conditions under which an Order Asymmetry Emerges

In the completed-utterance baseline, sequential context updating generated a mean size-first advantage of approximately 1.1 percentage points, while context-fixed semantics yielded no difference between the orders (Figure 8). With fixed meanings, the two adjective orders have the same completed interpretation and therefore the same target probability. Updating breaks this symmetry because colour restricts the comparison class for size. This baseline recovers the proposed communicative contribution of hierarchical restriction.

Including successive adjective information changed the result. With only the two orders available, the global production model yielded a mean colour-first advantage under both semantic regimes: approximately 2.7 percentage points with fixed meanings and 1.8 with updating. The first adjectives can differ in informativeness even when the completed meanings coincide, allowing an ordering asymmetry under fixed semantics. Its direction, however, depends on the alternatives available for reference. Adding the single-adjective responses changed the mean advantage to size-first under both regimes, to approximately 2.3 and 2.8 percentage points, respectively. The speaker can now identify a referent with one adjective, changing how probability is distributed over the two-adjective descriptions and how the pragmatic listener interprets their use. Adjective selection and utterance length thus affect the communicative value of an order even when the two orders themselves remain unchanged.

Expanding to fifteen utterances retained the mean size-first advantage, at approximately 2.6 percentage points with fixed semantics and 3.1 with updating. At the semantic constants used in production, both regimes yielded almost identical advantages of approximately 1.9 percentage points. Updating's additional contribution therefore depends on the semantic settings, while the broader utterance space supports a size-first advantage under either regime.

The simulation shows how the interpretation and selection of adjectives jointly determine the communicative advantage of an order. Semantic updating can generate an advantage when comparing completed meanings; evaluating successive adjective information and allowing shorter descriptions changes both the source and direction of that advantage. The availability of single-adjective utterances is consequential for the behaviour of the same two-adjective orders. This result provides a computational motivation for the paper's central empirical choice: to examine adjective order within the full distribution of adjective selection, utterance length, and order.

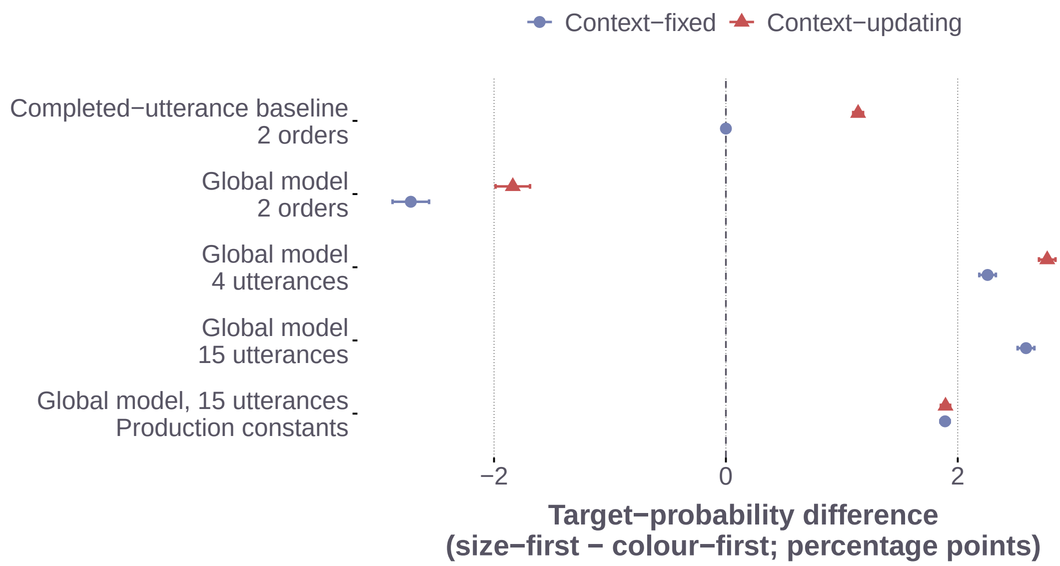

Figure 8: Controlled changes in the size-first pragmatic target-recovery advantage on the same 24,000 random scenes. The completed-utterance baseline scores only the final literal-listener target probability. The remaining rows use the global production model ($\kappa = 0$), which accumulates listener log probabilities over successive adjectives. The four-utterance space adds single adjectives to the two orders; the fifteen-utterance space contains all production responses. The first four rows average over the semantic parameter grid; the final row uses the production semantic constants. Points show scene-averaged target-probability differences; positive values favour size-first and negative values favour colour-first. Intervals are ± 1.96 Monte Carlo standard errors, calculated after averaging parameter settings within scenes.

7 General Discussion

The behavioural studies place adjective ordering within the broader problem of reference generation. Explicit slider judgements exhibit a replicated context gradient: size-first preference is strongest when size identifies the target, weakest when colour does, and remains slightly positive in colour-sufficient scenes. Production distributes these contextual demands across adjective selection, utterance length, and initial position. Speakers often select one sufficient adjective, and lower size discriminability increases redundant adjective use most clearly when size alone is sufficient. Higher discriminability favours size-initial responses most clearly when both properties are needed. The effect of perceptual reliability therefore depends on the referential demands of the scene and on which aspect of the description is considered.

These findings connect ordering preferences to the decisions that determine which adjectives are produced together. When one adjective resolves reference, an explicit preference between two completed orders may have little opportunity to affect the utterance produced. When speakers include additional properties, the resulting descriptions introduce further competition over both content and order. The full production distribution consequently provides evidence about the mechanisms of adjective ordering through choices about what to mention and how much to say.

7.1 Plan-Guided Production

With participant-specific variation in pragmatic optimality and stable order expectations controlled, plan-guided incremental production outpredicts both global and fully incremental production. The comparison supports a joint contribution of complete-utterance evaluation and competition among successive adjective choices. The clearest architecture differences occur in form-involving displays, where colour and form compete for the initial position. These responses extend the evidence beyond the motivating size–colour contrast: the influence of successive competition becomes visible when several properties and continuations remain available.

The architecture result connects the usefulness of a completed description to the alternatives encountered as that description develops. An adjective contributes to the referential value of the whole utterance while competing with other properties that could be mentioned at the same position. The fitted plan-guided model allows both contributions to shape production. This account accords with the broader possibility that an utterance-level plan constrains production while local alternatives continue to influence successive word choices (Ferreira and Swets, 2002; Konopka and Meyer, 2014). The production comparison provides a distributional basis for examining how these contributions are implemented during planning.

7.2 Semantic Composition and Ordering Advantages

Context-fixed semantics predicts the complete response distribution more accurately than sequential context updating within hierarchical semantic composition. The predictive difference is concentrated in adjective content and length, with an uncertain aggregate conditional-order contribution. The semantic comparison therefore bears on how interpretations affect the selection of a description as a whole. Evaluating only the relative frequency of the two size–colour orders would leave much of this difference unobserved.

The simulation clarifies how the proposed communicative benefit of semantic updating depends on the production choices under consideration. Updating creates a size-first advantage when the two orders are evaluated through their completed meanings, consistent with the contribution proposed for hierarchical restriction (Franke, Scontras, and Simonic, 2019). Once successive adjective information enters the global model's utility, the two-order comparison instead favours colour-first on average under both semantic regimes. Allowing single-adjective alternatives reverses this average direction, and the broader production space sustains a size-first advantage even with fixed meanings. The meaning of a completed description therefore supplies only part of its communicative value: the listener also reasons about why the speaker chose it from the available alternatives. This connects the simulation directly to the observed prevalence of single-adjective responses and the joint modelling of adjective selection, length, and order. The fitted advantage of context-fixed semantics can thus coexist with a communicative ordering advantage; predicting speakers' complete choices provides evidence about the contribution of updating in the present task.

7.3 Participant Variation and Production Planning

Allowing the contribution of successive adjective choice to vary across participants further improves prediction, particularly for multi-adjective utterances. These longer descriptions involve several successive choices and can include properties beyond those needed for unique reference. Participant variation consequently helps explain which multi-adjective utterances speakers produce.

Redundant production can reflect the early encoding of accessible properties or the communicative value of additional information for visual search and uncertainty reduction (Pechmann, 1989; Engelhardt, Bailey, and Ferreira, 2006; Rubio-Fernández, 2016; Degen et al., 2020). These pressures can converge in a shared visual task, where a property that is readily available to the speaker may also be useful to the listener. The fitted response patterns provide targets for experiments that independently manipulate property accessibility, partner knowledge, and listener uncertainty. Planning-time manipulations and online measures such as speech onset and eye movements can further examine when later adjectives become available and how their availability affects earlier choices (Ferreira and Swets, 2002; Konopka and Meyer, 2014). Such evidence would connect the present account of completed utterances to the time course of reference production.

7.4 Generalisation Beyond the Present Task

The experiments use German prenominal noun phrases in a controlled reference task with a fixed adjective vocabulary. Model development used these data, and response-wise leave-one-out prediction evaluates the resulting selected specifications. The present evidence motivates a further test on new participants and materials, including freer production and designs that independently manipulate lexical subjectivity and perceptual reliability. These extensions would examine how the relation between complete-utterance evaluation and successive adjective choice generalises across referential settings.

The results locate adjective order within choices about what to mention and how much to say. Plan-guided incremental production provides the best predictions by combining complete-utterance evaluation with competition among successive adjective choices. The resulting account connects stable ordering preferences to the context-sensitive decisions that determine which adjectives are produced in the first place.

8 Data Availability Statement

Data, analysis scripts, model code, final posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the deterministic production input, the final global, plan-guided, and fully incremental model artifacts, preprocessing and reconstruction receipts, and the scripts required to reproduce the reported behavioural and predictive analyses.

9 Author Note

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A Supplementary Experimental Methods and Results

Table 1 summarises the participant exclusions and the observations included in each analysis. Participants could meet more than one exclusion criterion; the table therefore reports both criterion-specific counts and the total number excluded. For Study 2, the production model comparison reported in Section 5 uses all 9,100 annotated observations across the fifteen utterance categories and three adjective-pair families. The behavioural analyses focus on the size–colour subset.

Table 1: Participant exclusions and analysis-table sizes. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Analysis table used in manuscript
Study 1a slider ratings	$150 \to 134$	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three "none fits" responses on experimental items; 16 unique participants excluded.	Broad cleaned critical table: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	$132 \to 104$	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	Cleaned replication table: 8,379 critical observations. Reported size–colour slider subset: 2,786 observations.
Production study	$149 \to 113$	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Canonical full-data table: 9,100 annotated observations. Reported size–colour behavioural subset: 3,034 observations.

The slider stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The stored large-size band is $\{9, 10\}$ in both groups; the smaller-size band is $\{1, 2, 3\}$ at high discriminability and $\{1, 2, 3, 4, 5, 6\}$ at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials. The released stimulus table specifies the six object sizes, colours, and forms for every list–item–condition combination.

The slider-rating materials used the fixed adjective vocabulary in Table 2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table 2: Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels `1--6`, `9`, `10`	*großen*, *kleinen*
Colour	`blue`, `orange`, `grey`, `brown`, `green`, `black`	*blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen*
Form	`circle`, `diamond`, `heart`, `quadrat`, `star`, `triangle`	*runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen*

For modelling, each response is matched to its recorded display. Colour and form are represented by whether each object matches the target on that property; the lexical forms shown to participants are listed above.

The first author coded production responses into the 15 ordered adjective categories listed in Table 3. The data files use `D` for the size or dimension adjective. The model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 3: Production coding dictionary. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.

Length	Codes
One adjective	`D`, `C`, `F`
Two adjectives	`DC`, `DF`, `CD`, `CF`, `FD`, `FC`
Three adjectives	`DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD`

A.1 Behavioural Analysis Specification

The behavioural hypotheses were evaluated with Bayesian hierarchical models. All analyses used the size–colour subsets reported in the main experiment figures: 3,485 observations in Study 1a, 2,786 in Study 1b, and 3,034 in Study 2. The later production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete fifteen-category response space and all 9,100 production trials. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \text{Normal}(\mu_{ij}, \sigma), \qquad \mu_{ij} = \mathbf{x}_{ij}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ij}^{\mathsf{T}}\mathbf{b}_i, \qquad \mathbf{b}_i \sim \text{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{participant}}).$$

For production, size-initiality indicated whether the coded response began with the size adjective. Over-informativeness indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k:

$$y_{ijk} \sim \mathrm{Bernoulli}(\pi_{ijk}), \qquad \mathrm{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ijk}^{\mathsf{T}}\mathbf{b}_i + c_k, \qquad c_k \sim \mathrm{Normal}(0, \sigma_{\mathrm{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in the two one-property-sufficient contexts. For over-informativeness, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\mathrm{Normal}(0, .30)$, $\mathrm{HalfNormal}(.20)$ priors on participant standard deviations, and a $\mathrm{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\mathrm{Normal}(0, 2)$ intercept prior, $\mathrm{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\mathrm{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\mathrm{LKJ}(2)$ prior. Four chains run in parallel each provided $1{,}000$ retained draws after $1{,}000$ warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\widehat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned contrasts were then evaluated on the rating or response-probability scale.

A.2 Additional stimulus and procedure details

Critical scenes were realised through systematic permutations of colour and form values applied to 27 six-object scene templates. This varied the surface displays while preserving the referential structure of each condition. Each study began with three practice trials with feedback. Main trials were divided into four blocks of approximately equal length, with self-paced breaks and independently randomised trial order within each block. Study 1a participants completed 81 critical trials and 99 filler or control trials (180 total), with a mean completion time of 27.6 minutes. Control trials had an unambiguous expected response; the additional "none fits" option allowed participants to indicate that neither description identified the target. Frequent use of this option on experimental trials was an exclusion criterion, as detailed in Table 1. The cleaned Study 1b file contains the same 1,080 list–item–condition design tuples as Study 1a.

Study 2 participants completed 81 critical and 54 filler or control trials (135 total), with a mean completion time of 32.4 minutes. Fewer filler and control trials were used because typed responses were slower and more effortful than slider judgements. Compensation followed the same rate as Study 1a. Restricting responses to the adjective slot reduced off-task variability and provided a controlled basis for coding. The categorical response variable for the computational comparison is the ordered utterance label $y_i \in \{1, \ldots, 15\}$. For each binary behavioural outcome, matching responses were coded as 1 and all other annotated size–colour responses as 0.

A.3 Detailed behavioural results

Tables 4 and 5 report posterior medians and 95% credible intervals for the contrasts described in the main text. Slider estimates are in centred-rating units; production estimates are differences in response probability. The column P reports the posterior probability of the direction stated in each row; a dash indicates that a directional probability was not separately reported for that contrast in the main analysis summary. Interaction estimates compare effects across contexts, whereas within-context contrasts compare the two discriminability groups in one context.

The posterior probability of the graded context order was greater than .999 in Study 1a and .999 in Study 1b. Observed mean centred ratings in Study 1b were .272 in size-sufficient, .197 in both-necessary, and .054 in colour-sufficient contexts. Figure 9 shows the observed Study 1a ratings; the main-text figure shows fitted cell expectations with 95% credible intervals for both slider studies.

Observed redundancy proportions in size-sufficient displays were .90 under low and .63 under high size discriminability. Observed size-initial proportions in both-necessary displays were .49 under low and .65 under high

Table 4: Slider-rating contrasts, with interactions followed by context contrasts and within-context discriminability effects. Discriminability effects are high minus low. Interaction rows subtract the size-sufficient discriminability effect from the effect in the named context.

Study	Contrast	Median	95% CrI	P (direction)
1a	Interaction: both-necessary	$-.085$	$[-.151, -.022]$	.995 (<0)
1a	Interaction: colour-sufficient	$-.063$	$[-.172, .039]$	.885 (<0)
1a	Size-sufficient minus both-necessary	.094	$[.061, .127]$	—
1a	Both-necessary minus colour-sufficient	.171	$[.131, .208]$	—
1a	Colour-sufficient rating relative to zero	.054	$[.003, .109]$	.982 (>0)
1a	Discriminability: size-sufficient	.042	$[-.013, .099]$	—
1a	Discriminability: both-necessary	$-.043$	$[-.118, .028]$	—
1a	Discriminability: colour-sufficient	$-.021$	$[-.129, .082]$	—
1b	Interaction: both-necessary	$-.014$	$[-.097, .071]$	.623 (<0)
1b	Interaction: colour-sufficient	.034	$[-.082, .157]$	.715 (>0)
1b	Size-sufficient minus both-necessary	.073	$[.029, .115]$	—
1b	Both-necessary minus colour-sufficient	.140	$[.090, .190]$	—
1b	Colour-sufficient rating relative to zero	.057	$[.001, .116]$	.977 (>0)
1b	Discriminability: size-sufficient	.090	$[.014, .160]$	—
1b	Discriminability: both-necessary	.076	$[-.011, .161]$	—
1b	Discriminability: colour-sufficient	.124	$[.016, .240]$	—

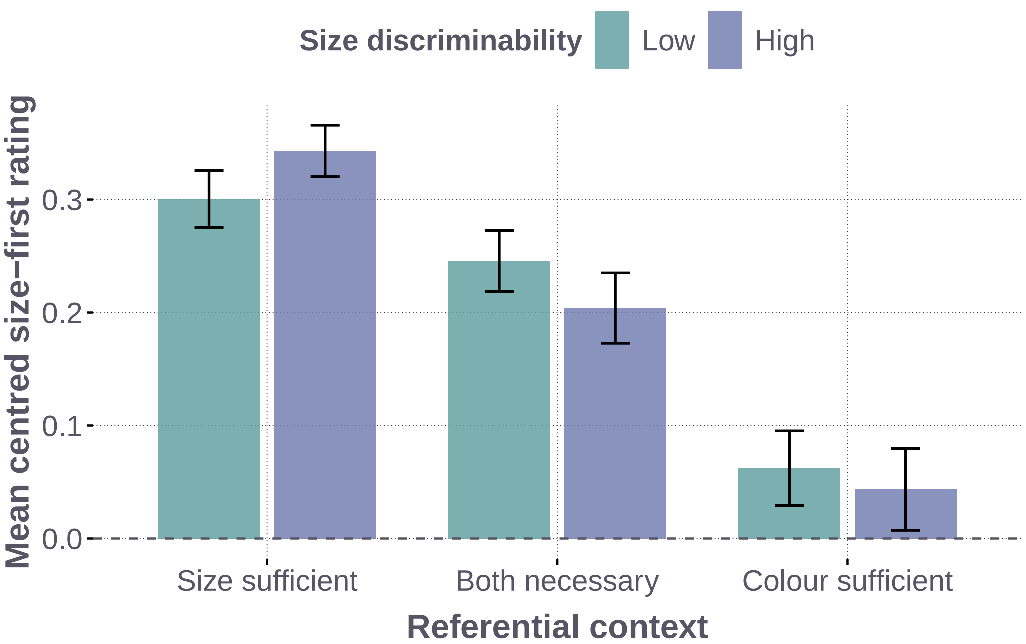

Figure 9: Observed Study 1a slider ratings. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

Table 5: Production contrasts on the response-probability scale. The redundancy interaction compares the low-minus-high difference in size-sufficient displays with its mean in the other two contexts. The size-initiality interaction compares the high-minus-low difference in both-necessary displays with its mean in the two one-property-sufficient contexts.

Contrast	Median	95% CrI	P (direction)
Redundancy: planned interaction	.153	$[.072, .236]$	$> .999$ (> 0)
Redundancy: size-sufficient, low minus high	.234	$[.151, .318]$	$> .999$ (> 0)
Size-initiality: planned interaction	.169	$[.077, .256]$	$> .999$ (> 0)
Size-initiality: both-necessary, high minus low	.141	$[.036, .242]$	.997 (> 0)
Size-initiality: size-sufficient, high minus low	$-.021$	$[-.101, .057]$	—
Size-initiality: colour-sufficient, high minus low	$-.032$	$[-.068, -.006]$	—

discriminability. These descriptive proportions differ from the posterior-predictive contrasts in Table 5, which integrate over participant and item variation.

A descriptive check of the complete production data also showed group differences in the colour–form family. In form-sufficient displays, form-initial responses were more frequent in the high- than in the low-discriminability group (.77 vs. .71). Because discriminability was manipulated between participants, these differences can also reflect group variation in how the task was approached. The production model represents participant variation alongside the visual manipulation.

B Production Model Specification

B.1 Representation of Experimental Displays

Participant list, condition, and item identifiers link each retained response to its recorded six-object display. The model represents each display as a 6×3 array of size, colour, and form values, with the intended referent indexed first. This representation preserves the observed values of all three properties, including those outside the adjective pair defining the trial.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the comparison class $\mathcal{C}$ (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta\left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C})\right), \tag{9}$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o, graded membership is

$$m_{\mathrm{S}}(o \mid \mathcal{C}) = \Phi\left(\frac{s_o - \theta_\delta(\mathcal{C})}{w\sqrt{s_o^2 + \theta_\delta(\mathcal{C})^2 + \epsilon}}\right), \qquad w = .6856, \quad \epsilon = 10^{-8}, \tag{10}$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1 - \nu$, with $\nu_{\mathrm{C}} = .59$ and $\nu_{\mathrm{F}} = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (1), (2), (4) and (5) define how this meaning enters the literal listener L and the three production accounts.

B.3 Component Inventory

Table 6: Final production-model components.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Graded support for matching colour
Form reliability	Fixed neutral	Form production supported by the task link
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Task-link coefficients	Estimated	Support for sufficient single adjectives, form use, and length and discriminability effects
Successive-choice weight κ_i	Estimated in plan-guided production	Shared weight or participant-specific weights for successive adjective choice

The population pragmatic-optimality parameter has prior $\alpha_0 \sim \mathrm{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log\beta} \sim \mathrm{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log\alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log\beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log\alpha} \sim \mathrm{HalfNormal}(.15)$, and $\tau_\beta \sim \mathrm{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \mathrm{Uniform}(0, 1)$. In the participant-weight refinement, $\mu_\kappa \sim \mathrm{Uniform}(0, 1)$ and $\kappa_i = \mathrm{logit}^{-1}(\mathrm{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \mathrm{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \mathrm{Normal}(0, 1)$. The salience coefficient has a $\mathrm{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\mathrm{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\mathrm{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\mathrm{HalfNormal}(1)$ prior. Threshold selectivity is fixed at $\delta = .50$; colour reliability, form reliability, and perceptual blur take the values specified above. A final Laplace smoothing step mixes the predicted distribution with a uniform distribution over the fifteen responses, with a fixed weight of .003 on the uniform component.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the displayed scene. For the target o^*, let s_o, c_o, and f_o be an object's recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_\mathrm{S} = \frac{1+h}{2}\,\mathrm{logistic}\left(\frac{s_{o^*} - \max_{o \neq o^*} s_o}{\mathrm{sd}(s) + 10^{-8}}\right), \tag{11}$$

$$r_\mathrm{C} = 1 - \frac{1}{5}\sum_{o \neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \qquad r_\mathrm{F} = 1 - \frac{1}{5}\sum_{o \neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \tag{12}$$

Here $\mathrm{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_\mathrm{S}, b_\mathrm{C}, b_\mathrm{F}) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3}\sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_\mathrm{salience} z_a$ and the continuation cost in the task link below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u, the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|}\sum_{v \in \mathcal{U}_{A(u)}} \log q(v). \tag{13}$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Task link for adjective selection and length. The task link $T(u, X)$ connects the referential design to choices about which properties to mention and how far to continue a description. Let $n = |u|$ and $A(u)$ denote the length and unordered adjective set of u. For this term, X includes the display's condition labels: $d \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}, \varnothing\}$ is the property that identifies the target on its own, with $d = \varnothing$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the selected models use

$$\begin{aligned} T(u, X) = & -\rho_{\text{cont}} \sum_{t=1}^{n-1} \max(z_{w_t}, 0) + \lambda_{\text{suff}} \mathbf{1}\{d \neq \varnothing,\ u = (d)\} \\ & + \lambda_{\mathrm{F}}(1 - c)\mathbf{1}\{\mathrm{F} \in A(u)\} - \lambda_3 \mathbf{1}\{n = 3\} \\ & + \lambda_{\mathrm{S}} h\, \mathbf{1}\{d = \mathrm{S},\ u = (\mathrm{S})\} \\ & + \lambda_{\mathrm{SF}}(1 - 2h)\mathbf{1}\{d = \mathrm{S},\ A(u) = \{\mathrm{S}, \mathrm{F}\}\}. \end{aligned} \tag{14}$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (2). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing descriptions outside colour-sufficient conditions. With form reliability fixed at .50, form gives equal literal support to matching and nonmatching objects; this task term provides a direct contribution to form use. The three-adjective penalty offsets the value accumulated along longer descriptions. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Numerical evaluation and response likelihood. The implementation uses 64-bit arithmetic and a floor of 10^{-8} for size memberships, listener target probabilities before taking logarithms, and local choice probabilities. Colour and form memberships receive an additive 10^{-8} offset. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of floored local log-choice probabilities, and $B(u) = \beta_i \Omega(u) + T(u, X)$, the computed score before Laplace smoothing is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\widetilde{S}_{\kappa_i}(u \mid X)$, whose endpoints obey the geometric interpolation in Equation (6). The categorical response likelihood is

$$p(y_{it} = u \mid X_{it}) = (1 - .003)\widetilde{S}_{\kappa_i}(u \mid X_{it}) + \frac{.003}{15}. \tag{15}$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ; the final refinement estimates κ_i. All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

B.4 Hierarchical Inference Structure

Figure 10 separates fixed theoretical choices, estimated parameters, participant variation, and observed responses. The fixed variables A and R index production architecture and semantic regime. For participant i on trial t, the model maps the display and target X_{it} and the stable-order score $\Omega(u)$ to a probability $\pi_{it}(u)$ over the fifteen possible utterances, from which the observed response y_{it} is drawn. The shared coefficients govern visual salience, adjective selection, and utterance length.

C Production Model Comparison and Diagnostics

C.1 Inference and Predictive Comparison

Each likelihood evaluation scores all fifteen completed utterances and performs the required semantic composition and prefix-level normalisation. To make this exact calculation feasible, we precomputed trial-independent

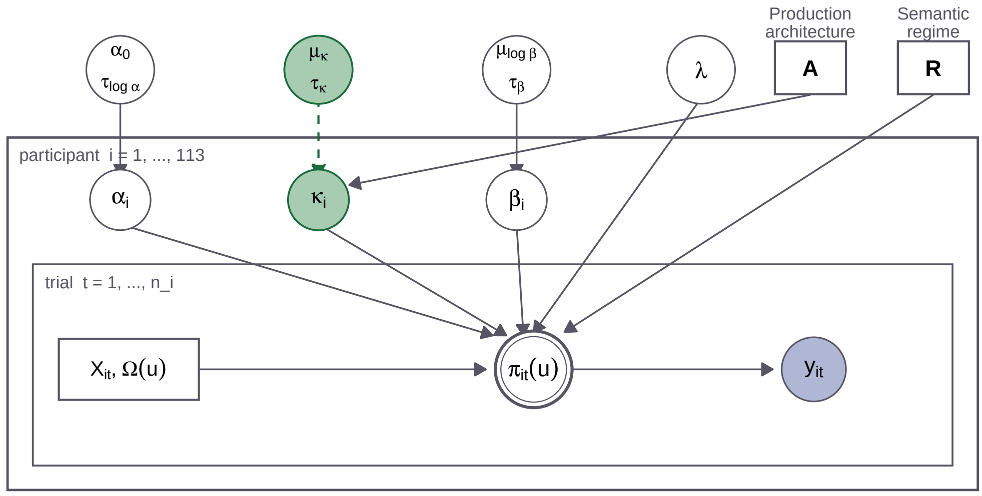

Figure 10: Formal plate for the final joint hierarchical model and the restrictions used in the matched comparisons. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The dashed κ hierarchy is activated only in the final plan-guided refinement.

utterance and prefix structures and vectorised and compiled the listener and speaker computations in JAX. All eight production fits used four No-U-Turn Sampler (NUTS) chains on a GPU and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8 to obtain adequate mixing; the other six models retained 1,000 draws per chain at depth 6. The selected fits had zero divergences, maximum $\widehat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum Bayesian fraction of missing information (BFMI) above .64, and no observation with Pareto $k > .7$. The participant-weight plan-guided fits reached the depth-6 step cap on 77.7% of transitions under fixed semantics and 31.7% under updating; this limits sampling efficiency despite the passing numerical diagnostics. The best-fitted context-fixed model had an absolute ELPD of $-13{,}039.924$, maximum $\widehat{R} = 1.0051$, and minimum BFMI of $.742$. Independent semantic checks verified that each modifier is applied once from the noun outwards and that changing the semantic regime can change predictions at matched parameters. Endpoint, interpolation, and direct-versus-precomputed probability checks agreed within numerical precision, and posterior likelihood replay agreed with each stored artifact within 10^{-8}.

The pointwise leave-one-out score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is an additional completed response from a sampled participant, as described in Section 5.1.

Table 7: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy.

Contrast	ΔELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ, minus average endpoints	402.677	$[311.466, 503.098]$	$> .999$
Fully incremental minus global	-291.596	$[-684.673, -1.321]$	.024
Updating minus fixed, shared-κ grid	-43.527	$[-56.005, -31.770]$	$< .001$
Participant κ_i hierarchy, context-fixed	364.742	$[248.178, 520.967]$	$> .999$
Participant κ_i hierarchy, updating	372.285	$[255.597, 526.056]$	$> .999$

Averaging over the two semantic regimes, the plan-guided model had the highest predictive accuracy of the three architectures in .994 of the Bayesian-bootstrap draws. In the context-fixed plan-guided model, the posterior mean shared successive-choice weight was

$$\kappa = .446, \qquad 95\% \text{ credible interval} = [.407, .486]. \tag{16}$$

C.2 Architecture-Level Response Patterns

Figure 11 shows initial-adjective cells in which the largest difference among architecture predictions is at least 6.7 percentage points and traces these differences into completed utterances. Colour-initial responses contributed 328.912 ELPD to the matched plan-guided model's advantage over the average endpoint, or 81.7% of the summed pointwise contrast. Across the displayed cells, plan-guided production often lies between the global and fully incremental endpoints. The response partitions locate this gain by summing exact-response log scores within the observed categories.

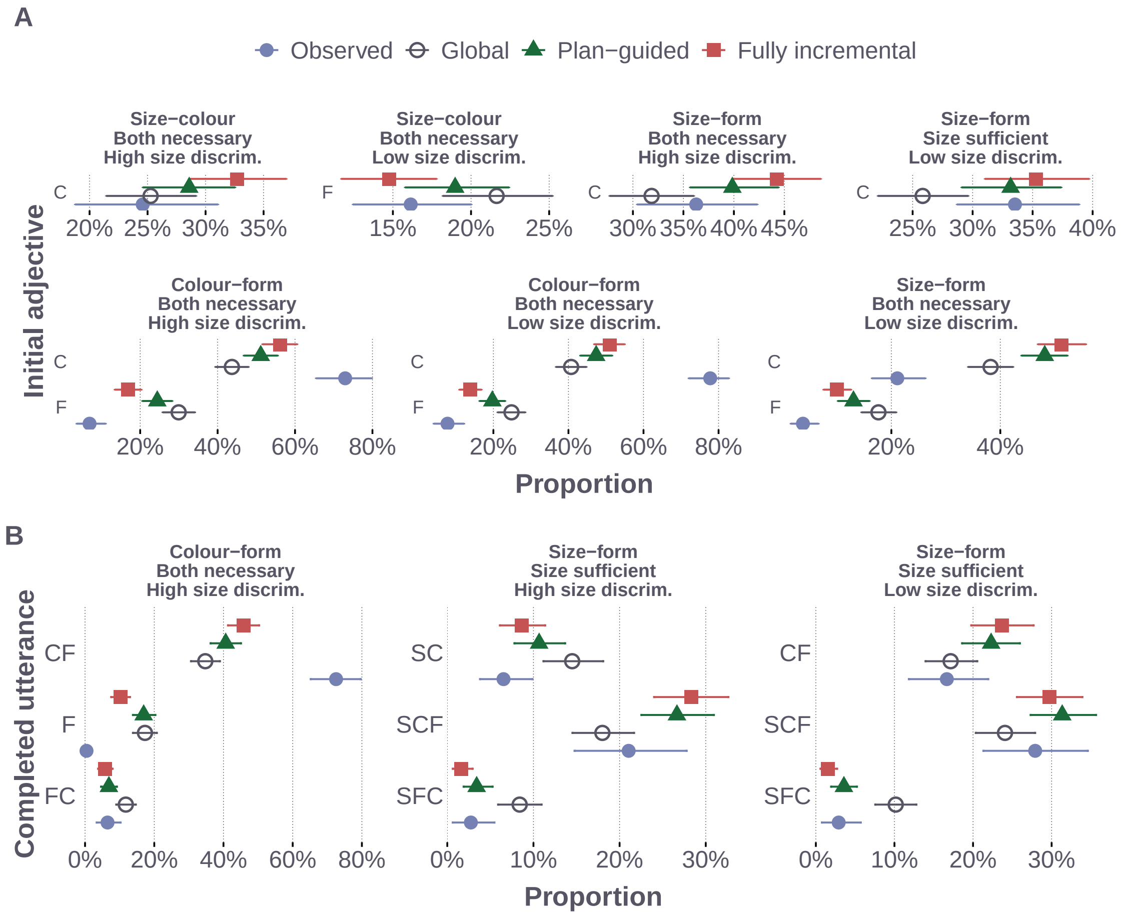

Figure 11: Qualitative comparison of the matched production architectures. Panel A shows initial-adjective cells for which the maximum difference among model predictions is at least 6.7 percentage points. Panel B shows completed utterances with differences of at least five percentage points within the three experimental conditions with the largest pairwise total-variation distance among model predictions. Response labels use S for size, C for colour, and F for form. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Horizontal scales vary across conditions to make the comparatively small differences among production architectures visible. The displayed cells are descriptive diagnostics; the held-out comparison in Figure 6 provides the evidence for relative predictive accuracy.

C.3 Decomposing the Semantic Predictive Contrast

The architecture-specific updating-minus-fixed effects were -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible interval excluded zero. The sequentially updating participant-specific model was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$.

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-it}(u \mid X_{it}) = p_{-it}(\ell(u) \mid X_{it})\, p_{-it}(A(u) \mid \ell(u), X_{it})\, p_{-it}(u \mid A(u), X_{it}). \quad (17)$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. All three contributions refer to the same held-out response. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws using seed 20260906. For the participant-weight plan-guided pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the participant-weight pair, the original response contrast sums to -17.548 ELPD in size–colour trials, -51.172 in size–form trials, and 14.580 in colour–form trials. These partitions locate the predictive difference within the complete response distribution.

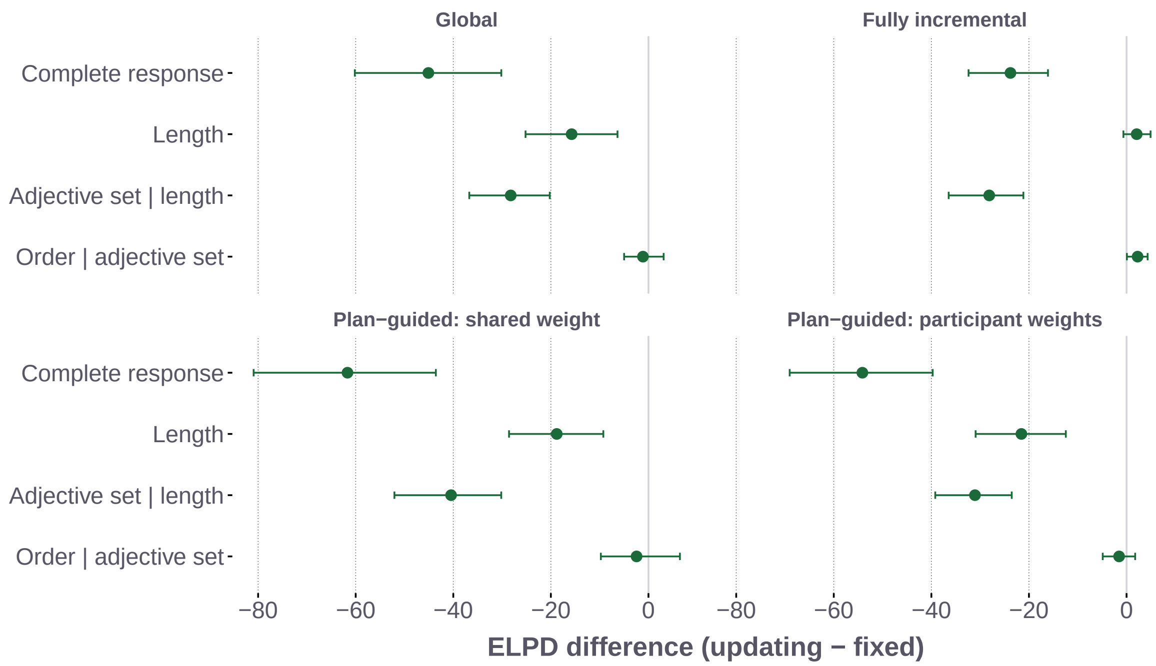

Figure 12: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific successive-choice weights; the other plan-guided panel uses a shared weight.

C.4 Posterior-Predictive Adequacy

Across the 270 cells formed by the fifteen completed utterances, three adjective families, three referential contexts, and two size-discriminability conditions, the posterior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points. The corresponding 95% posterior-predictive intervals contained 58.5% of the observed cell proportions, a descriptive check of fit across the dependent response cells.

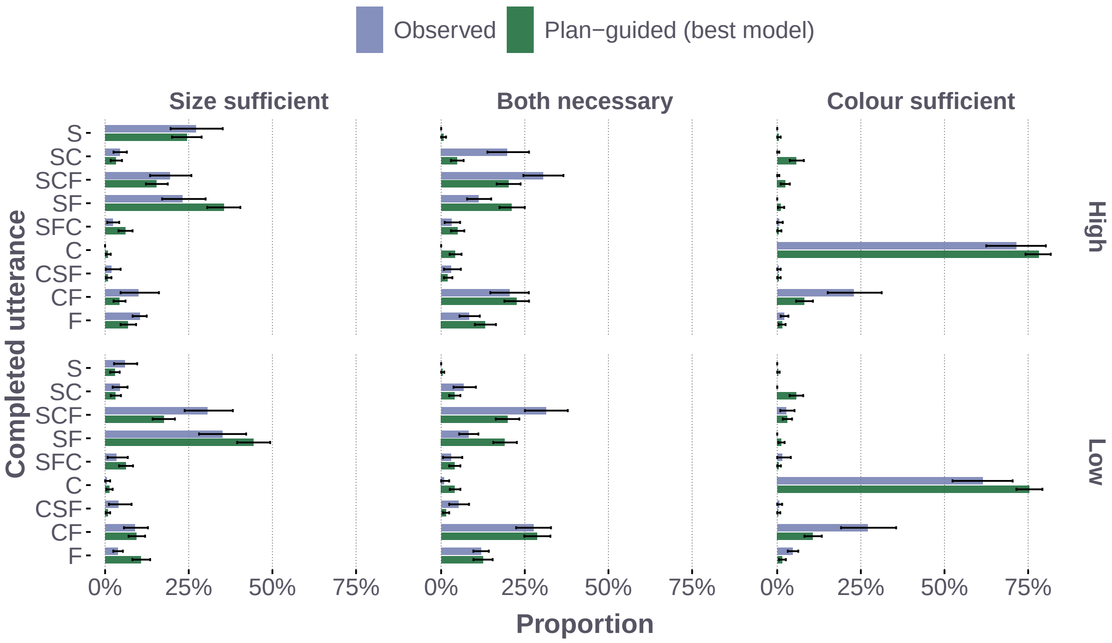

Figure 13: Observed and posterior-predicted proportions for frequent utterances in size–colour trials under the best-fitted plan-guided model. The figure includes utterances whose observed or mean posterior-predicted proportion reaches 5% in at least one context-by-discriminability condition, with proportions pooled across trials within each condition. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Adjective labels preserve surface order: S denotes size, C colour, and F form.

D Model-Development Checks

Model development used the present data to select the form semantics, stable-order feature, and task terms. The matched comparisons hold these components constant across production architectures and semantic regimes.

Form semantics and use. Form reliability is fixed at .50, and an estimated task coefficient supports form use. Earlier variants used informative form meanings tied to colour reliability and alternative priors on the form-use coefficient.

Stable order source. The selected specification uses the GPT-2 order residual. Earlier variants added a term for reliability-based order preferences or omitted this score.

Task link. The selected specification includes sufficient-single-adjective support, form use, a three-adjective penalty, and size-discriminability terms. An earlier reduced-policy variant removed these terms as a bundle. These exploratory variants were evaluated before the semantic implementation was corrected and were not refitted. They informed model development; the predictive comparisons reported here use the common specification in Appendix B.

E Participant Variation and Predictive Contributions

The final joint model estimates participant-specific pragmatic optimality, successive-choice weight, and stable-order weight simultaneously. Table 7 reports its predictive gain relative to the otherwise matched shared-κ model under each semantic regime. In the best-fitted context-fixed model, the population location of the participant-specific successive-choice weights was $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$.

Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of $[.525, .763]$ and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval $[163.012, 319.334]$, and 183.796 ELPD on three-adjective utterances, 95% credible interval $[100.757, 304.826]$. Its contribution to one-adjective utterances was -48.426 ELPD (95% credible interval $[-70.371, -28.666]$).

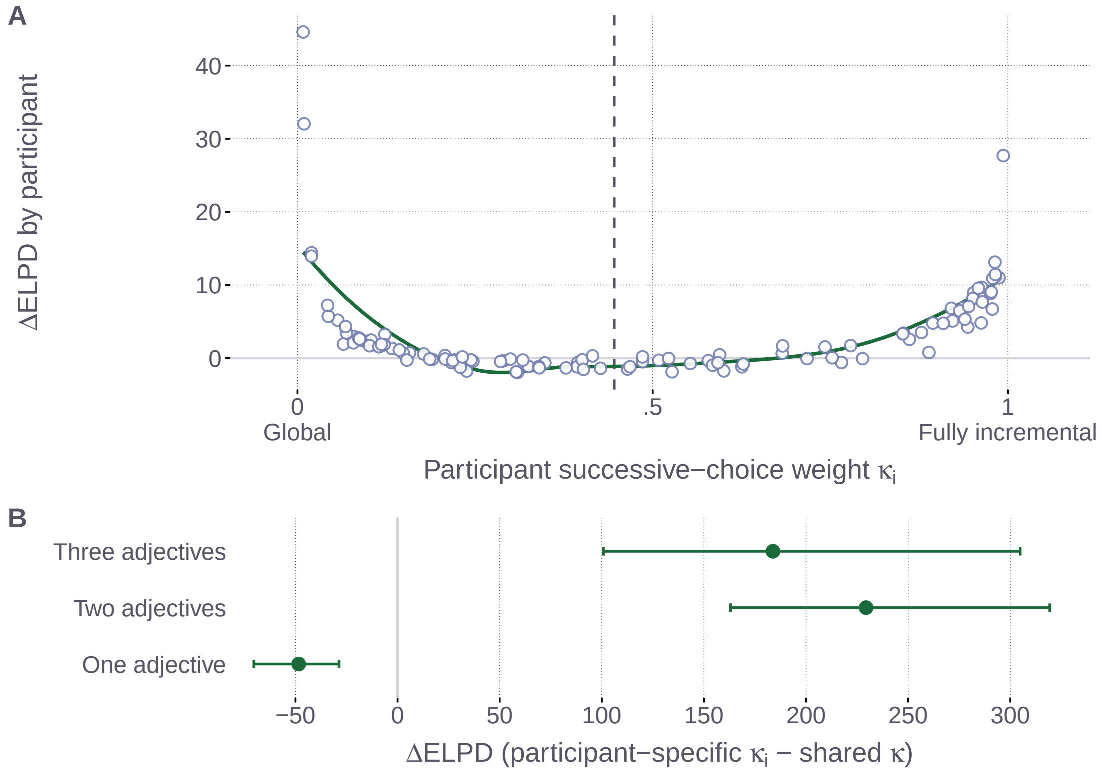

Figure 14: Where participant-specific successive-choice weights improve prediction. Panel A relates each participant's leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i; the dashed line marks the shared-κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-fitted plan-guided model and its otherwise matched shared-κ model to localise the held-out model contrast.

Posterior-predictive contrasts between participants in the lower and upper tails of each parameter localise their fitted response signatures. Higher α_i concentrates probability on a sufficient single adjective, especially S in size-sufficient displays and F in form-sufficient displays. Higher κ_i mainly redistributes probability among form-involving responses whose continuation sets differ after the first adjective. Higher β_i shifts probability from single-adjective responses towards longer responses that follow the stable order score. In the context-fixed joint model, the mean participant-level posterior correlations were .010 for α_i with κ_i and .025 for α_i with β_i. The posterior participant-level association between κ_i and β_i was positive, with mean $r = .162$ and 95% credible interval $[.073, .249]$. These correlations and tail contrasts describe how the three parameters organise response variation within the joint fit.

Maximal three-adjective utterances accounted for 1,633 of the 9,100 modelled responses, or 17.9%. Within the fitted model, these responses balance accumulated referential value and salience against a shared three-adjective cost, with α_i, κ_i, and β_i determining how that balance is expressed for each participant. These quantities jointly determine the predicted frequency and ordering of three-adjective utterances.

F Local Normalisation and Model Endpoints

Local adjective competition. For the utilities defined in Equation (2), taking the logarithm of the product of locally normalised choices gives the accumulated utility minus the local-normalisation term. Suppressing the shared referent and scene arguments,

$$\log \prod_t \frac{\exp U_t(w_t \mid u_{<t})}{\sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t})} = \sum_t U_t(w_t \mid u_{<t}) - \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}) \\ = U^{\text{complete}}(u) - \mathcal{N}^{\text{local}}(u). \tag{18}$$

Relation between the endpoints. At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The stable-order score, task link, and utterance space are identical across the three architectures. Numerical checks found agreement in category probabilities and log probabilities within a tolerance of 10^{-10}.

G Sequential Context Updating for Size

For the hierarchical semantic path, let C_{j-1} be the context state after the syntactically inner modifiers have composed. The semantic regime determines the comparison class used for size:

$$[\![\mathrm{S}]\!]_{\text{fix}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(\mathcal{C})}(o), \tag{19a}$$

$$[\![\mathrm{S}]\!]_{\text{upd},C_{j-1}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(C_{j-1})}(o). \tag{19b}$$

where θ_δ derives a graded size threshold from a comparison class and $\mathcal{C}$ is the original scene, represented by the uniform prior P_0. The context-updating listener recomputes the size threshold from the hierarchically prior state; the context-fixed listener anchors it to the scene prior. At each surface position, the production calculation evaluates the current prefix using this noun-outward interpretation. Each prefix is interpreted from the original object prior, with each modifier applied once.

H Simulation Parameter Sweeps

Each random display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to $[1, 30]$. The largest object is designated as the target and assigned blue; each non-target object is independently assigned blue or non-blue with equal probability. Three levels of contextual size spread vary how distinctly the largest target differs from its competitors. The semantic grid varies colour reliability, size-threshold selectivity, and perceptual blur, which control the strength of colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. In the simulation notation, k corresponds to threshold selectivity δ and w_f to perceptual blur w in Equations (9) and (10).

Figures 16 to 18 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, X)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b, the global and incremental speakers are

$$S_G(ab \mid o, X) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \tag{20}$$

$$R_I(ab \mid o, X) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \tag{21}$$

$$S_I(ab \mid o, X) = \frac{R_I(ab \mid o, X)}{R_I(ab \mid o, X) + R_I(ba \mid o, X)}. \tag{22}$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The pragmatic listener uses a uniform referent prior, giving $L^{\text{prag}}(o \mid ab, X) = S(ab \mid o, X) / \sum_r S(ab \mid r, X)$. The implementation clips very small intermediate probabilities at 10^{-20}.

The reproducible sweep uses seed 20260906 and samples 1,000 independent scenes for each context-size and size-spread combination. Each scene is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent scenes and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within scenes; repeated evaluations of one scene do not count as independent samples. The computations use CUDA and 64-bit arithmetic.

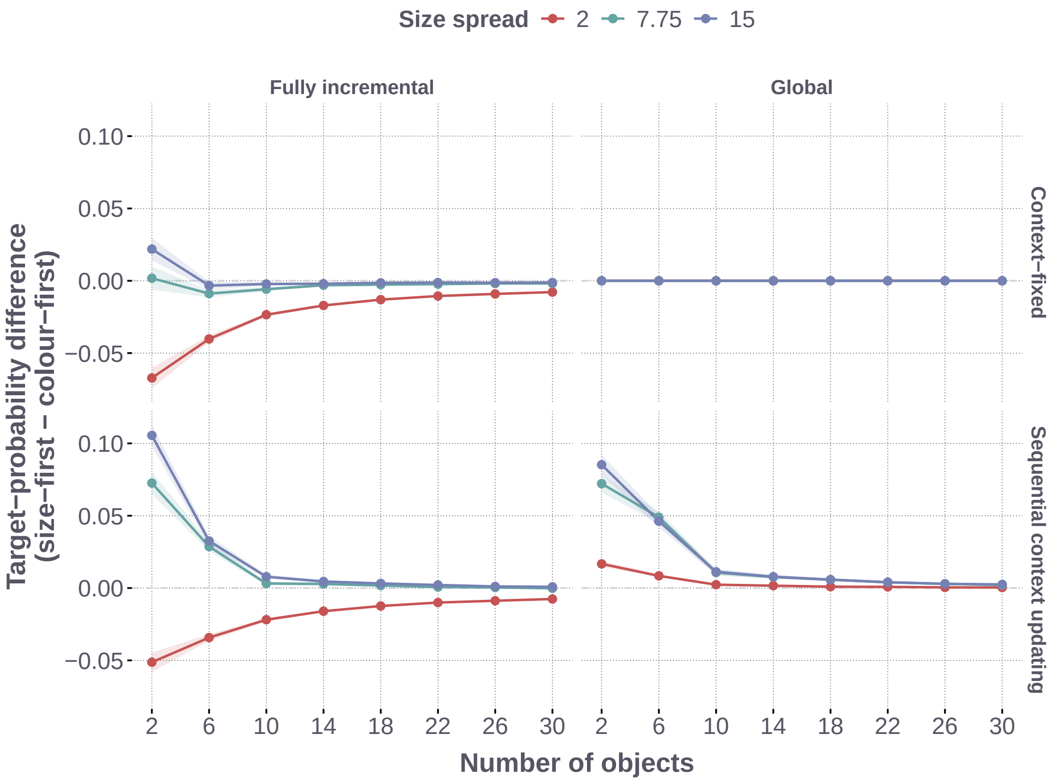

Figure 15: Size-first advantage Δ_{SF} across context size and contextual spread, shown for the global and fully incremental speakers under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled scene. The complete parameter grid appears in Table 8. Positive values indicate a higher model-assigned target probability for the size-first order.

Figure 15 shows how the size-first advantage changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Wider size distributions generally make the largest target more distinguishable from its competitors; clustered sizes reduce this distinction.

The simulation's global context-fixed model provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first advantage under global production. The advantage is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

Fully incremental production adds a pressure from local informativity along the surface sequence. When sizes are closely clustered, colour is the more informative opening cue and the incremental speaker favours colour-first. As sizes spread apart, size becomes a more reliable opening cue and the colour-first pressure weakens. After averaging over the semantic parameter settings in Figure 15, colour-first reversals occur under fully incremental production; the global context-fixed model remains symmetric, and the global context-updating model remains size-first. Individual parameter settings can yield small negative estimated advantages under global updating, so the average pattern does not establish a uniform advantage throughout the grid. Under incremental production, sequential context updating shifts the balance towards size-first when the colour-restricted comparison class changes size interpretation sufficiently strongly. The parameter sweeps in Figures 16 to 18 show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

Table 8: Simulation parameter grid

Parameter	**Grid values**	**# levels**
Speaker architecture	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
Colour reliability ν	0.90, 0.92, 0.94, 0.96, 0.98	5
Threshold selectivity k	0.10, 0.30, 0.50, 0.70, 0.90	5
Perceptual blur w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Scenes per context-size/spread pair	1,000	
Total model evaluations	**12,000,000**	

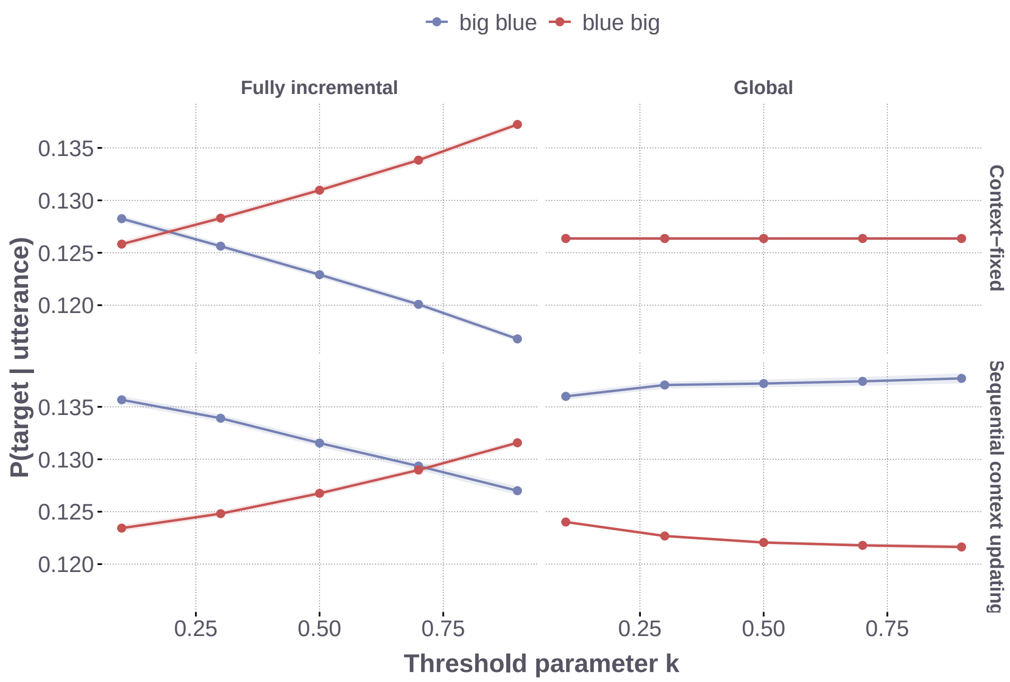

Figure 16: Effect of the threshold parameter k on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

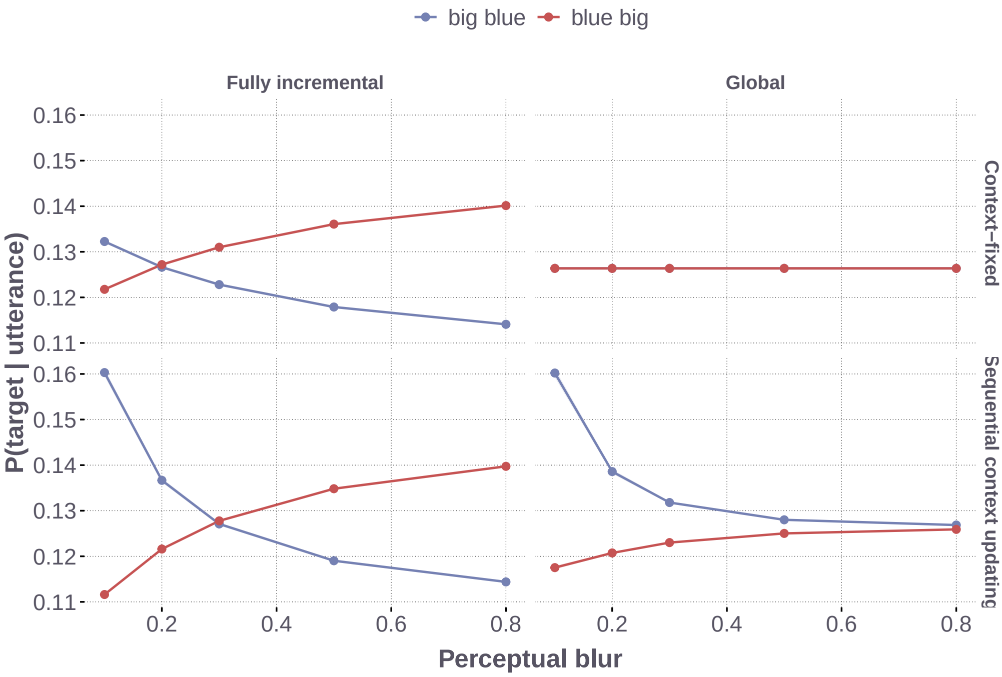

Figure 17: Effect of perceptual blur w_f on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

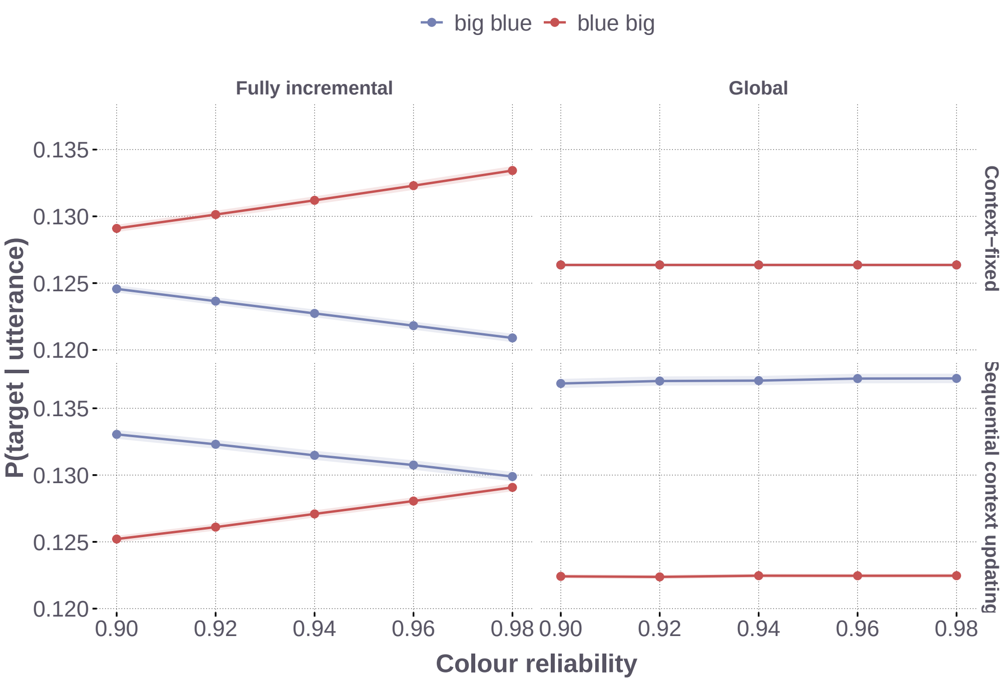

Figure 18: Effect of colour reliability ν on simulated pragmatic target recovery, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

H.1 Controlled Changes on the Same Random Scenes

The completed-utterance control, labelled terminal evaluation below, uses a different utility from the global production model:

$$U_{\mathrm{term}}(u, o, X) = \alpha \log L(o \mid u, X), \tag{23}$$

$$U_{\mathrm{prefix}}(u, o, X) = \alpha \sum_{t=1}^{|u|} \log L(o \mid u_{1:t}, X). \tag{24}$$

Both scores are globally normalised over completed utterances. The prefix score with $\kappa = 0$ is the global model used in the main text; $\kappa = .5$ and $\kappa = 1$ give the plan-guided and fully incremental simulation models. The terminal score supplies a control for the additional contribution of intermediate adjective information.

The controlled comparisons reuse all 24,000 scenes from the original sweep and the target-recovery measure in Equation (8). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order's probability under the full speaker distribution, without first conditioning on the two orders. The comparisons include terminal evaluation and prefix evaluation at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. The full fifteen-response implementation agrees with the production forward calculation with task, salience, and order terms set to zero and final Laplace smoothing omitted. This isolates the computational components without assigning fitted participant effects or experimental task labels to random scenes.

The main comparison holds pragmatic optimality at $\alpha = 1$ and excludes stable-order expectations, salience, task-specific response terms, and final Laplace smoothing. Form reliability is fixed at .50. The main figure isolates successive adjective information and utterance-space changes while holding global choice fixed. Its four global-model rows use $\kappa = 0$; the full architecture comparison additionally includes fixed weights $\kappa = .5$ and $\kappa = 1$ for plan-guided and fully incremental production. The production constants are colour reliability .59, threshold selectivity .50, and perceptual blur .6856. Table 9 reports the mean advantages shown in Figure 8; its prefix-evaluation rows use the global model ($\kappa = 0$). For the fifteen-response global model at the production constants, the paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$.

Table 9: Mean size-first pragmatic target-recovery advantage in percentage points on the same 24,000 random scenes.

Evaluation	Responses	Semantic settings	Context-fixed	Updating
Terminal	2	Parameter grid	0.000	1.141
Prefix	2	Parameter grid	−2.718	−1.838
Prefix	4	Parameter grid	2.257	2.772
Prefix	15	Parameter grid	2.589	3.104
Prefix	15	Production constants	1.891	1.894

The main figure uses the production semantic implementation throughout. As a separate implementation control, substituting its lexical calculation into the original two-order global rule reduces the mean updating advantage from 1.465 to 1.142 percentage points across the original grid. The terminal control additionally retains the production listener log-probability floor, giving the 1.141-point advantage in Table 9. The implementations differ in probability floors and numerical normalisation around weighted quantiles; the control measures their combined impact. The original curves are retained in Figure 15.

Figure 19 reports the fifteen-response comparison across evaluation rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This representative setting comes from the fixed model's population log-weight location; it does not integrate posterior or participant uncertainty. At $\kappa = 0$, adding this term gives mean advantages of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive advantages are already present without this term. All other salience and task terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each scene before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups. These averages summarise the specified parameter grid; individual contexts can favour different orders. Reproduction checks recovered the stored two-order outputs, and the controlled probabilities agreed with the corresponding production calculation to numerical precision.

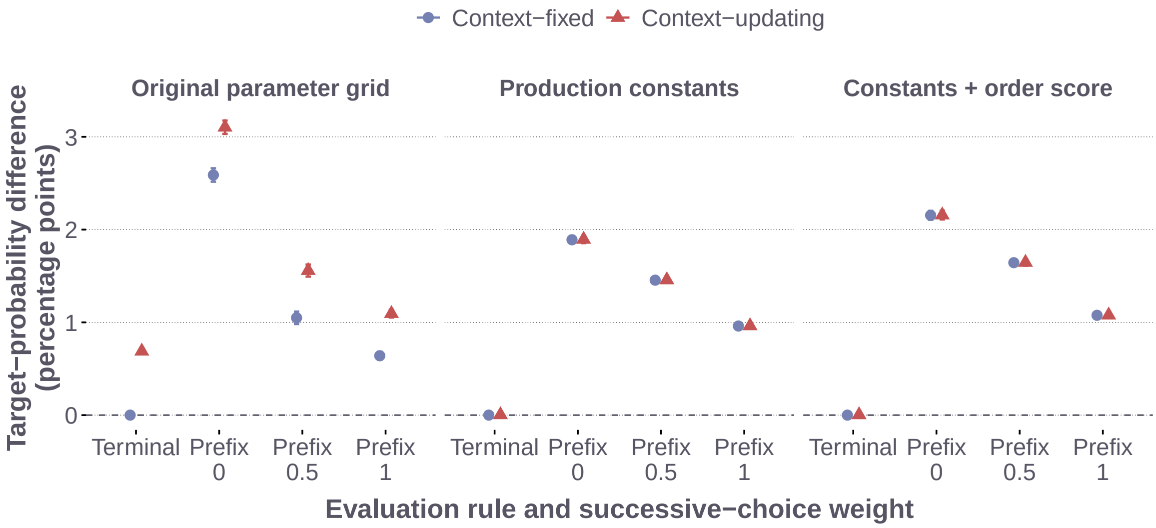

Figure 19: Controlled fifteen-response comparison across evaluation rules, semantic parameters, and a shared stable-order score. Terminal evaluation uses only the completed-utterance score. Prefix evaluation accumulates prefix scores with local-normalisation weight $\kappa = 0, .5, 1$. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 scene-level Monte Carlo standard errors. All comparisons use the same random scenes; no participant posterior uncertainty is represented.

H.2 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the 3,034 size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and task link. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote. Because the production responses were coded by adjective type, fixing these named adjective extensions across candidate referents is an additional assumption of the listener calculation. This is a model-based extension to counterfactual referents within the observed task context.

Let $S_{it}^{(d)}(u \mid o, X)$ be the resulting complete-response probability at posterior draw d. We evaluate the speaker's size-first probability conditional on the size–colour adjective set as

$$\frac{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X)}{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X) + S_{it}^{(d)}(\mathrm{CS} \mid o^*, X)}. \tag{25}$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval. Replaying the original target reproduces its stored likelihood within 10^{-8} for every model.

Figure 20 reports the resulting speaker and listener predictions. In the participant-weight plan-guided models, the mean size-first target-probability advantage was .0062 under fixed semantics, 95% posterior interval $[.0047, .0077]$, and .0058 under updating, $[.0045, .0072]$. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were .967 and .966. Both fitted regimes therefore generate a size-first advantage under this counterfactual listener calculation. These posterior-derived predictions, together with the response-score decomposition, show how a simulated ordering advantage can coexist with lower predictive accuracy for human adjective content and length.

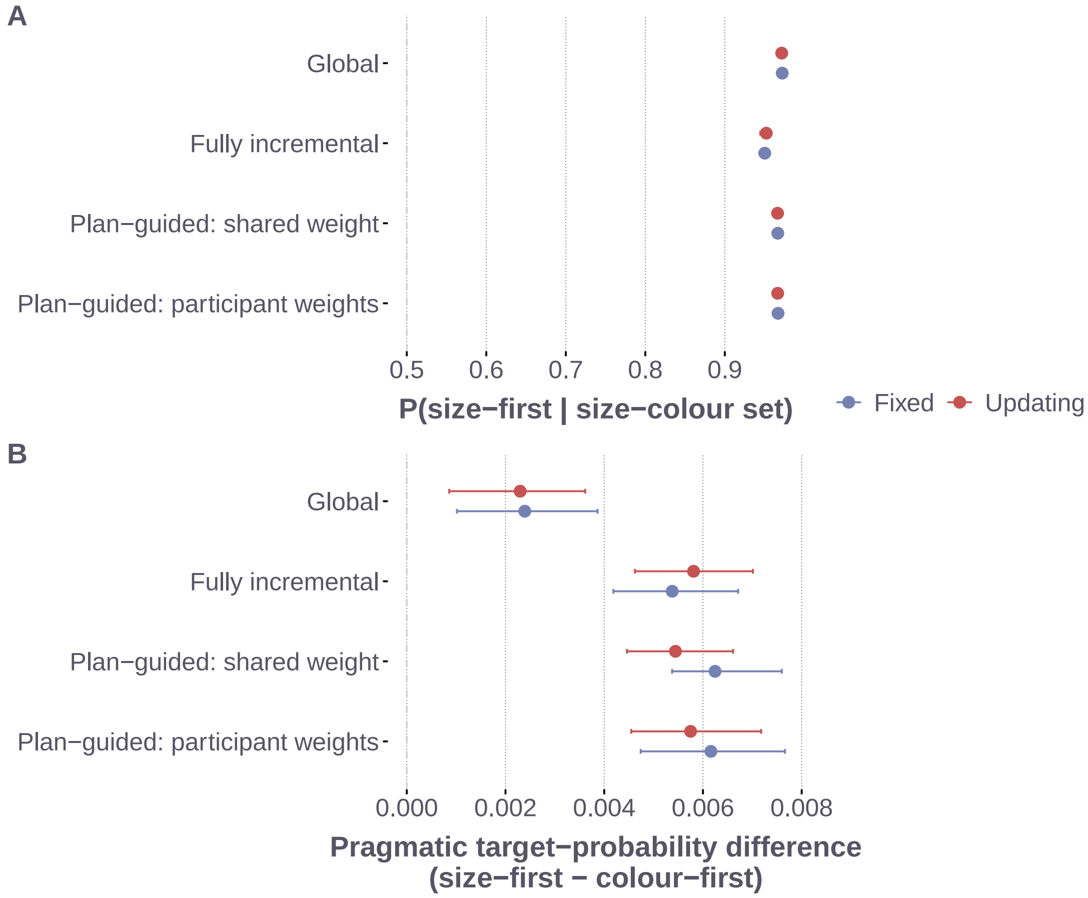

Figure 20: Supplementary predictions from the fitted production models on the size–colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size–colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 posterior parameter draws per model.